

brane. Consequently, blockage of the membrane led to lower rates of productivity and/or the need for filter aids to be added, and these were serious disadvantages.

In contrast, an alternative which is rapidly gaining prominence both in the processing of whole fermentation broths (Tanny *et al.*, 1980; Brown and Kavanagh, 1987; Warren *et al.*, 1991) and cell lysates (Gabler and Ryan, 1985; Le and Atkinson, 1985) is cross-flow filtration. Here, the flow of medium to be filtered is tangential to the membrane (Fig. 10.14(a)), and no filter cake builds up on the membrane.

The benefits of cross-flow filtration are:

- (a) Efficient separation, > 99.9% cell retention.
- (b) Closed system; for the containment of organisms with no aerosol formation (see also Chapter 7).
- (c) Separation is independent of cell and media densities, in contrast to centrifugation.
- (d) No addition of filter aid (Zahka and Leahy, 1985).

The major components of a cross-flow filtration system are a media storage tank (or the fermenter), a pump and a membrane pack (Fig. 10.14(b)). The membrane is usually in a cassette pack of hollow fibres or flat sheets in a plate and frame type stack or a spiral cartridge (Strathmann, 1985). In this way, and by the

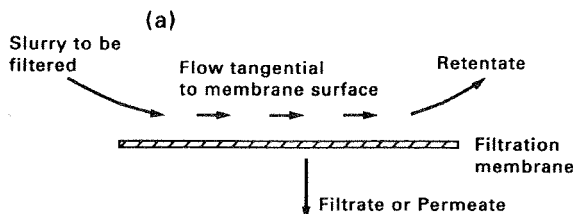


FIG. 10.14a. Schematic diagram of cross-flow filtration.

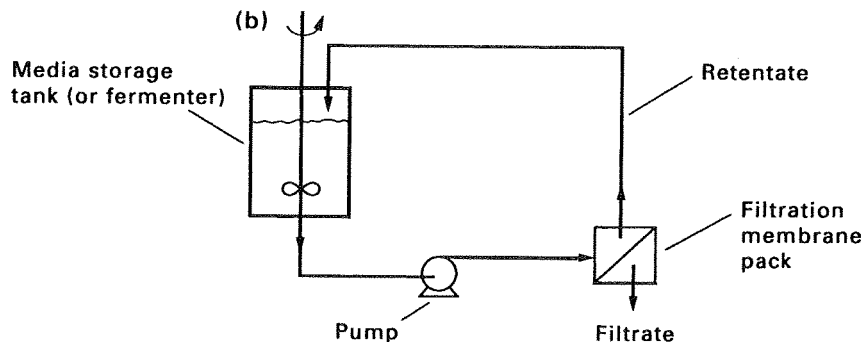


FIG. 10.14b. Major components of a cross-flow filtration system.

introduction of a much convoluted surface, large filtration areas can be attained in compact devices. Two types of membrane may be used; microporous membranes with a specific pore size (0.45, 0.22 μm etc.) or an ultrafiltration membrane (see later section) with a specified molecular weight cut-off (MWCO). The type of membrane chosen is carefully matched to the product being harvested, with microporous and 100,000 MWCO membranes being used in cell separations.

The output from the pump is forced across the membrane surface; most of this flow sweeps the membrane, returning retained species back to the storage tank and generally less than 10% of the flow passes through the membrane (permeate). As this process is continued the cells, or other retained species are concentrated to between 5 and 10% of their initial volume. More complex variants of the process can allow *in-situ* washing of the retentate and enclosed systems for containment and sterilization (Mouro *et al.*, 1989).

Many factors influence filtration rate. Increased pressure drop will, up to a point increase flow across the membrane, but it should be remembered that the system is based on a swept clean membrane. Therefore, if the pressure drop is too great the membrane may become blocked. The filtration rate is therefore influenced by the rate of tangential flow across the membrane; by increasing the shear forces at the membrane's surface retained species are more effectively removed, thereby increasing filtration rate. Higher temperatures will increase filtration rate by lowering the viscosity of the media, though this is clearly of limited application in biological systems. Filtration rate is inversely proportional to concentration, and media constituents can influence filtration rate in three ways. Low molecular weight compounds increase media viscosity and high molecular weight compounds decrease shear at the membrane surface, both leading to a reduction in filtration rate. Finally, broth constituents can 'foul' the

membrane, primarily by adsorption onto the membrane's surface, causing a rapid loss in efficiency. This can be controlled by modification of the membrane or media formulation in particular by reducing the use of antifoaming agents. Lee *et al.* (1993) have shown that the pulses of air injected into the flow to a cross-flow filter increase the shear rate at the membrane surface reducing the effects of membrane fouling.

CENTRIFUGATION

Micro-organisms and other similar sized particles can be removed from a broth by using a centrifuge when filtration is not a satisfactory separation method. Although a centrifuge may be expensive when compared with a filter it may be essential when:

1. Filtration is slow and difficult.
2. The cells or other suspended matter must be obtained free of filter aids.
3. Continuous separation to a high standard of hygiene is required.

Non-continuous centrifuges are of extremely limited capacity and therefore not suitable for large-scale separation. The centrifuges used in harvesting fermentation broths are all operated on a continuous or semi-continuous basis. Some centrifuges can be used for separating two immiscible liquids yielding a heavy phase and light phase liquid, as well as a solids fraction. They may also be used for the breaking of emulsions.

According to Stoke's law, the rate of sedimentation of spherical particles suspended in a fluid of Newtonian viscosity characteristics is proportional to the square of the diameter of the particles, thus the rate of sedimentation of a particle under gravitational force is:

$$V_g = \frac{d^2 g (\rho_P - \rho_L)}{18\mu} \quad (10.5)$$

where V_g = rate of sedimentation (m s^{-1})
 d = particle diameter (m)
 g = gravitational constant (m s^{-2})
 ρ_P = particle density (kg m^{-3})
 ρ_L = liquid density (kg m^{-3})
 μ = viscosity ($\text{kg m}^{-1} \text{s}^{-1}$)

This equation can then be modified for sedimentation in a centrifuge:

$$V_c = \frac{d \omega^2 r (\rho_P - \rho_L)}{18\mu} \quad (10.6)$$

where V_c = rate of sedimentation in the centrifuge (m s^{-1}),

ω = angular velocity of the rotor (s^{-1}),

r = radial position of the particle (m).

Dividing equation (10.6) by equation (10.5) yields

$$\frac{\omega^2 r}{g}$$

This is a measure of the separating power of a centrifuge compared with gravity settling. It is often referred to as the relative centrifugal force and given the symbol 'Z'.

It is evident from this formula that factors influencing the rate of sedimentation over which one has little or no control are the difference in density between the cells and the liquid (increased temperature would lower media density but is of little practical use with fermentation broths), the diameter of the cells (could be increased by coagulation/flocculation) and the viscosity of the liquid. Ideally, the cells should have a large diameter, there should be a large density difference between cell and liquid and the liquid should have a low viscosity. In practice, the cells are usually very small, of low density and are often suspended in viscous media. Thus it can be seen that the angular velocity and diameter of the centrifuge are the major factors to be considered when attempting to maximize the rate of sedimentation (and therefore throughput) of fermentation broths.

Cell aggregation and flocculation

Following an industrial fermentation it is quite common to add flocculating agents to the broth to aid de-watering (Wang, 1987). The use of flocculating agents is widely practised in the effluent-treatment industries for the removal of microbial cells and suspended colloidal matter (Delaine, 1983).

It is well known that aggregates of microbial cells, although they have the same density as the individual cells, will sediment faster because of the increased diameter of the particles (Stokes law). This sedimentation process may be achieved naturally with selected strains of brewing yeasts, particularly if the wort is chilled at the end of fermentation, and leads to a natural clearing of the beer.

Micro-organisms in solution are usually held as discrete units in three ways. Firstly, their surfaces are negatively charged and therefore repulse each other. Secondly, because of their generally hydrophilic cell walls a shell of bound water is associated with the cell

which acts as a thermodynamic barrier to aggregation. Finally, due to the irregular shapes of cell walls (at the macromolecular level) steric hindrance will also play a part.

During flocculation one or more mechanisms besides temperature can induce cell flocculation:

- (a) Neutralization of anionic charges, primarily carboxyl and phosphate groups, on the surfaces of the microbial cells, thus allowing the cells to aggregate. These include changes in the pH and the presence of a range of compounds which alter the ionic environment.
- (b) Reduction in surface hydrophilicity.
- (c) The use of high molecular weight polymer bridges. Anionic, non-ionic and cationic polymers can be used, though the former two also require the addition of a multivalent cation.

Flocculation usually involves the mixing of a process fluid with the flocculating agent under conditions of high shear in a stirred tank, although more compact and efficient devices have been proposed (Ashley, 1990). This stage is known as coagulation, and is usually followed by a period of gentle agitation when flocs developed initially are allowed to grow in size. The underlying theoretical principles of cell flocculation have been discussed by Atkinson and Daoud (1976).

Nakamura (1961) described the use of various compounds for flocculating bacteria, yeasts and algae, including alum, calcium salts and ferric salts. Other agents which are now used include tannic acid, titanium tetrachloride and cationic agents such as quaternary ammonium compounds, alkyl amines and alkyl pyridinium salts. Gasner and Wang (1970) reported a many hundred-fold increase in the sedimentation rate of *Candida intermedia* when recoveries of over 99% were readily obtained. They found that flocculation was very dependent on the choice of additive, dosage and conditions of floc formation, with the most effective agents being mineral colloids and polyelectrolytes. Nucleic acids, polysaccharides and proteins released from partly lysed cells may also bring about agglomeration. In SCP processes, phosphoric acid has been used as a flocculating agent since it can be used as a nutrient in medium recycle with considerable savings in water usage (Hamer, 1979).

The majority of flocculating agents currently in use are polyelectrolytes, which act by charge neutralization and hydrophobic interactions to link cells to each other. In processes where the addition of some toxic chemicals is to be avoided, alternative techniques have been

adopted. One method is to coagulate microbial protein which has been released from the cells by heating for short periods. Kurane (1990) reports the use of bioflocculants obtained from *Rhodococcus erythropolis*. They are suggested as being safer alternatives to conventional flocculants. Warne and Bowden (1987) suggest the use of genetic manipulation to alter cell surface properties to aid aggregation. Flocculating agents such as cross-linked cationic polymers may also be used in the processing of cell lysates and extracts prior to further downstream processing (Fletcher *et al.*, 1990). Bentham *et al.* (1990) utilized borax as a flocculating agent for yeast cell debris prior to decanter centrifugation.

The range of centrifuges

A number of centrifuges will be described which vary in their manner of liquid and solid discharge, their unloading speed and their relative maximum capacities. When choosing a centrifuge for a specific process it is important to ensure that the centrifuge will be able to perform the separation at the planned production rate, and operate reliably with minimum manpower. Large-scale tests may therefore be necessary with fermentation broths or other materials to check that the correct centrifuge is chosen.

THE BASKET CENTRIFUGE (PERFORATED-BOWL BASKET CENTRIFUGE)

Basket centrifuges are useful for separating mould mycelia or crystalline compounds. The centrifuge is most commonly used with a perforated bowl lined with a filter bag of nylon, cotton, etc. (Fig. 10.15). A continuous feed is used, and when the basket is filled with the filter cake it is possible to wash the cake before removing it. The bowl may suffer from blinding with soft biological materials so that high centrifugal forces cannot be used. These centrifuges are normally operated at speeds of up to 4000 rpm for feed rates of 50 to 300 dm³ min⁻¹ and have a solids holding capacity of 30 to 500 dm³. The basket centrifuge may be considered to be a centrifugal filter.

THE TUBULAR-BOWL CENTRIFUGE

This is a centrifuge to consider using for particle size ranges of 0.1 to 200 µm and up to 10% solids in the in-going slurry. Figure 10.16a shows an arrangement used in a Sharples Super-Centrifuge. The main component of the centrifuge is a cylindrical bowl (or rotor) (A in Fig. 10.16), which may be of a variable design de-

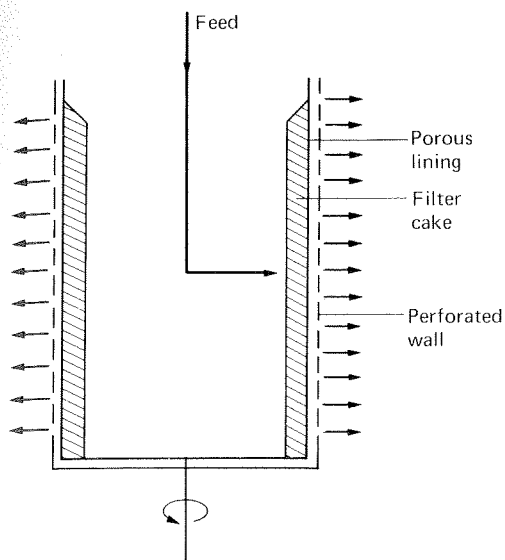


FIG. 10.15. Diagram of basket centrifuge.

pending on application, suspended by a flexible shaft (B), driven by an overhead motor or air turbine (C). The inlet to the bowl is via a nozzle attached to the bottom bearing (D). The feed which may consist of solids and light and heavy liquid phases is introduced by the nozzle (E). During operation solids sediment on the bowl wall while the liquids separate into the heavy phase in zone (G) and the light phase in the central zone (H). The two liquid phases are kept separate in their exit from the bowl by an adjustable ring, with the heavy phase flowing over the lip of the ring. Rings of various sizes may be fitted for the separation of liquids of various relative densities. Thus the centrifuge may be altered to use for:

- (a) Light-phase/heavy-phase liquid separation.
- (b) Solids/light-liquid phase/heavy-liquid phase separation.
- (c) Solids/liquid separation (using a different rotor, Fig. 10.16b).

The Sharples laboratory centrifuge with a bowl radius of approximately 2.25 cm can be operated with an air turbine at 50,000 rpm to produce a centrifugal force of approximately 62,000 g, but has a bowl capacity of only 200 cm³ with a throughput of 6 to 25 dm³ h⁻¹. The largest size rotor is the Sharples AS 26, which has a bowl radius of 5.5 cm and a capacity of 9 dm³, a solids capacity of 5 dm³ and a throughput of 390 to 2400 dm³ h⁻¹.

The advantages of this design of centrifuge are the

high centrifugal force, good dewatering and ease of cleaning. The disadvantages are limited solids capacity, difficulties in the recovery of collected solids, gradual loss in efficiency as the bowl fills, solids being dislodged from the walls as the bowl is slowing down and foaming. Plastic liners can be used in the bowls to help improve batch cycle time. Alternatively a spare bowl can be changed over in about 5 minutes.

THE SOLID-BOWL SCROLL CENTRIFUGE (DECANTER CENTRIFUGE)

This type of centrifuge is used for continuous handling of fermentation broths, cell lysates and coarse materials such as sewage sludge (Fig. 10.17). The slurry is fed through the spindle of an archimedean screw within the horizontal rotating solids bowl. Typically the speed differential between the bowl and the screw is in the range 0.5 to 100 rpm (Coulson and Richardson, 1991). The solids settling on the walls of the bowl are scraped to the conical end of the bowl. The slope of the cone helps to remove excess liquid from the solids before discharge. The liquid phase is discharged from the opposite end of the bowl. The speed of this type of centrifuge is limited to around 5000 rpm in larger models because of the lack of balance within the bowl, with smaller models having bowl speeds of up to 10000 rpm. Bowl diameters are normally between 0.2 and 1.5 metres, with the length being up to five times the diameter. Feed rates range from around 200 dm³ h⁻¹ to 200 m³ h⁻¹ depending on scale of operation and material being processed. A number of variants on the basic design are available:

- (a) Cake washing facilities (screen bowl decanters).
- (b) Vertical bowl decanters.
- (c) Facility for in-place cleaning.
- (d) Bio-hazard containment features; steam sterilization *in-situ*, two or three stage mechanical seals, control of aerosols, containment casings and the use of high pressure sterile gas in seals to prevent the release of micro-organisms.

THE MULTICHAMBER CENTRIFUGE

Ideally, this is a centrifuge for a slurry of up to 5% solids of particle size 0.1 to 200 μ m diameter. In the multichamber centrifuge (Fig. 10.18), a series of concentric chambers are mounted within the rotor chamber. The broth enters via the central spindle and then takes a circuitous route through the chambers. Solids collect on the outer faces of each chamber. The smaller

particles collect in the outer chambers where they are subjected to greater centrifugal forces (the greater the radial position of a particle, the greater the rate of sedimentation).

Although these vessels can have a greater solids capacity than tubular bowls and there is no loss of efficiency as the chamber fills with solids, their mechanical strength and design limits their speed to a maximum of 6500 rpm for a rotor 46-cm diameter with a holding capacity of up to 76 dm³. Because of the time needed to dismantle and recover the solids fraction, the size and number of vessels must be of the correct volume for the solids of a batch run.

THE DISC-BOWL CENTRIFUGE

This centrifuge relies for its efficiency on the pres-

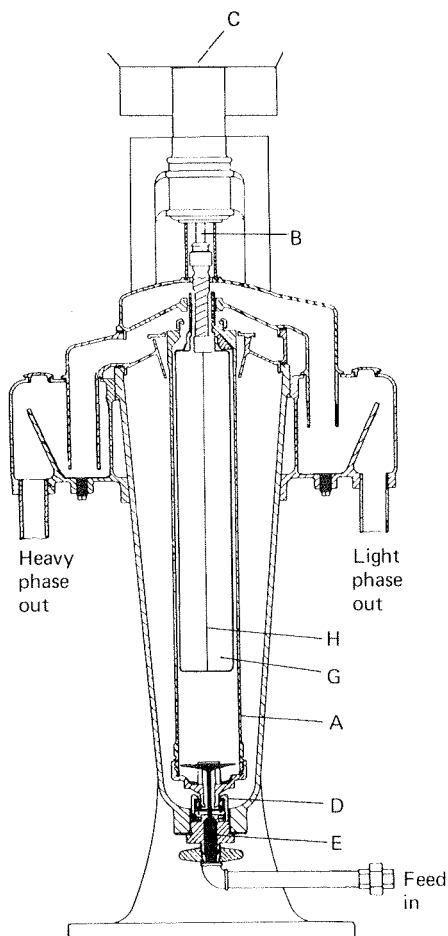


FIG. 10.16a. Section of a Sharples Super-Centrifuge (Alfa Laval Sharples, Camberley, U.K.).



FIG. 10.16b. A Sharples Super-Centrifuge assembled for discharge of one liquid phase (Alfa Laval Sharples, Camberley, U.K.).

ence of discs in the rotor or bowl (Fig. 10.19). A central inlet pipe is surrounded by a stack of stainless-steel conical discs. Each disc has spacers so that a stack can be built up. The broth to be separated flows outwards from the central feed pipe, then upwards and inwards between the discs at an angle of 45° to the axis of rotation. The close packing of the discs assists rapid sedimentation and the solids then slide to the edge of the bowl, provided that there are no gums or fats in the slurry, and eventually accumulate on the inner wall of the bowl. Ideally, the sediment should form a sludge which flows, rather than a hard particulate or lumpy sediment. The main advantages of these centrifuges are

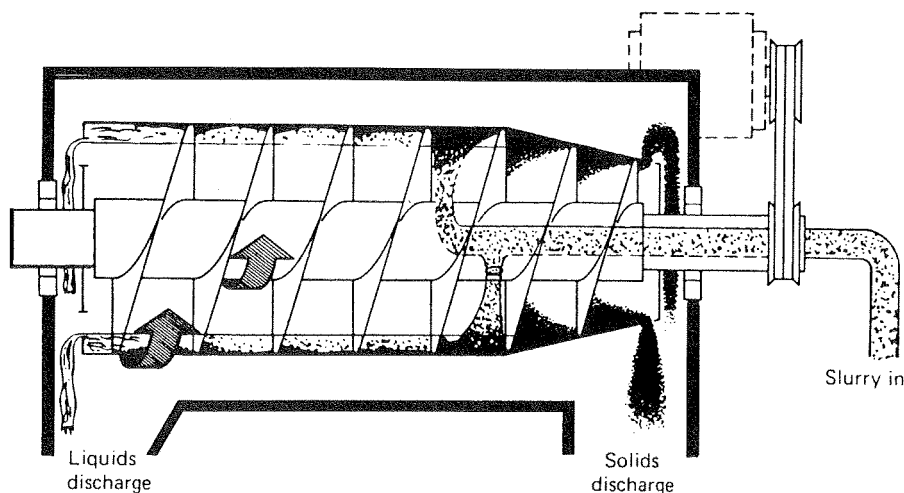


FIG. 10.17a. Diagram of a solid-bowl scroll centrifuge (Alfa Laval Sharples Ltd, Camberley, U.K.).

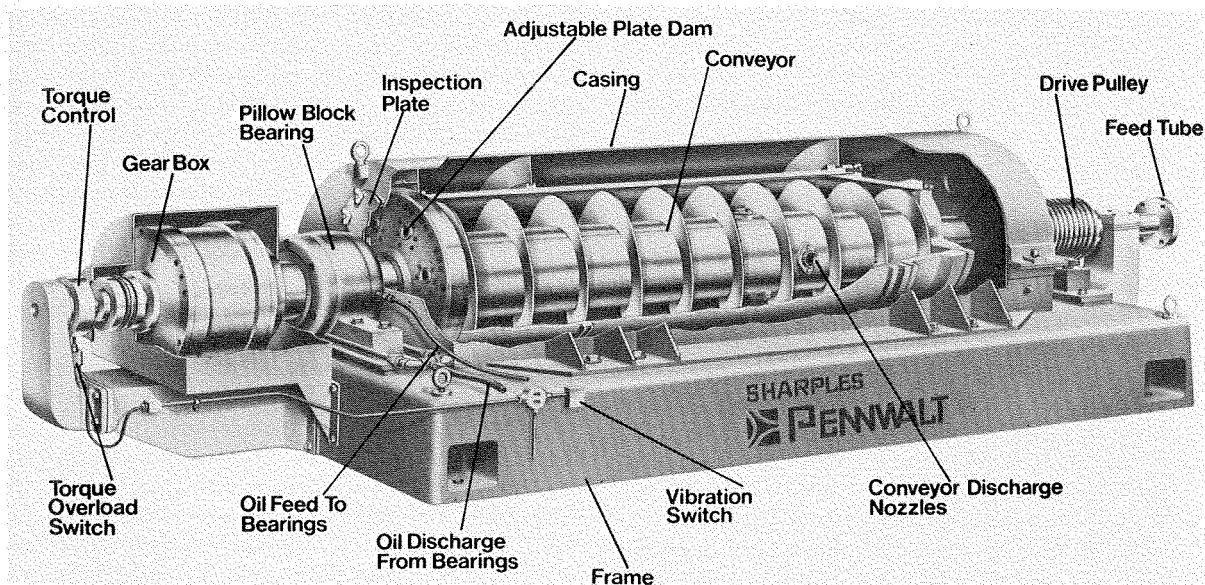


FIG. 10.17b. Cutaway view of a Sharples Super-D-Canter continuous solid-bowl centrifuge, Model P-5400 (Alfa Laval Sharples Ltd, Camberley, U.K.).

their small size compared with a bowl without discs for a given throughput. Some designs also have the facility for continuous solids removal through a series of nozzles in the circumference of the bowl or intermittent solids removal by automatic opening of the solids collection bowl. The arrangement of the discs makes this type of centrifuge laborious to clean. However, recent models such as the Alfa Laval BTUX 510 (Alfa Laval Sharples Ltd, Camberley, Surrey, U.K.) system (Fig.

10.20) are designed to allow for cleaning *in-situ*. In addition this and similar plant have the facility for *in-situ* steam sterilization and total containment, incorporating double seals to comply with containment regulations (see also Chapter 7). Feed rates range from 45 to 1800 dm³ min⁻¹, with rotational speeds typically between 5000 and 10,000 rpm. The Westfalia CSA 19-47-476 is also steam sterilizable and has been used for the sterile collection of organisms (Walker *et al.*,

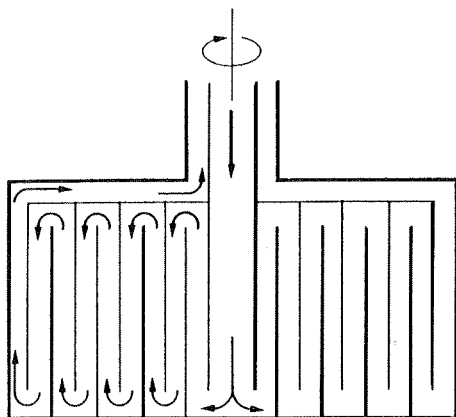


FIG. 10.18. L.S. of a multichamber centrifuge.

1987). Similarly, the Westfalia CSA 8 can be modified for contained operation and steam sterilization (Frude and Simpson, 1993).

CELL DISRUPTION

Micro-organisms are protected by extremely tough cell walls. In order to release their cellular contents a number of methods for cell disintegration have been developed (Wimpenny, 1967; Hughes *et al.*, 1971). Any potential method of disruption must ensure that labile materials are not denatured by the process or hydrolysed by enzymes present in the cell. Huang *et al.* (1991) report the use of a combination of different techniques to release products from specific locations within yeast cells. In this way the desired product can be obtained with minimum contamination. Although many techniques are available which are satisfactory at laboratory scale, only a limited number have been proved to be suitable for large-scale applications, particularly for intracellular enzyme extraction (Wang *et al.*, 1979; Darbyshire, 1981). Containment of cells can be difficult or costly to achieve in many of the methods described below and thus containment requirements will strongly influence process choice. Methods available fall into two major categories:

Physico-mechanical methods

- (a) Liquid shear.
- (b) Solid shear.
- (c) Agitation with abrasives.
- (d) Freeze-thawing.
- (e) Ultrasonication.

Chemical methods

- (a) Detergents.
- (b) Osmotic shock.
- (c) Alkali treatment.
- (d) Enzyme treatment.

Physical-mechanical methods

LIQUID SHEAR

Liquid shear is the method which has been most widely used in large scale enzyme purification procedures (Scawen *et al.*, 1980). High-pressure homogenizers used in the processing of milk and other products in the food industry have proved to be very effective for microbial cell disruption. One machine, the APV-Manton Gaulin-homogenizer (The APV Co. Ltd, Crawley, Surrey, U.K.), which is a high-pressure positive displacement pump, incorporates an adjustable valve with a restricted orifice (Fig. 10.21). The smallest model has one plunger, while there are several in larger models. During use, the microbial slurry passes through a non-return valve and impinges against the operative valve set at the selected operating pressure. The cells then pass through a narrow channel between the valve and an impact ring followed by a sudden pressure drop at the exit to the narrow orifice. The large pressure drop across the valve is believed to cause cavitation in the slurry and the shock waves so produced disrupt the cells. Brookman (1974) considered the size of the pressure drop to be very important in achieving effective disruption, and as with all mechanical methods, cell size and shape influence ease of disruption (Wase and Patel, 1985). The working pressures are extremely high. Hetherington *et al.* (1971) used a pressure of 550 kg cm⁻² for a 60% yeast suspension. A throughput of 6.4 kg soluble protein h⁻¹ with 90% disruption could be achieved with a small industrial machine. In larger models, flow rates of up to 600 dm³ h⁻¹ are now possible and operating pressures of 1200 bar are utilized in some processes (Asenjo, 1990). Darbyshire (1981) has stressed the need for cooling the slurry to between 0 and 4°C to minimize loss in enzyme activity because of heat generation during the process. The increase in slurry temperature is approximately proportional to the pressure drop across the valve. Because of problems caused by heat generation and because cell suspensions can be surprisingly abrasive, it is common practice to operate such homogenizers in a multi-pass

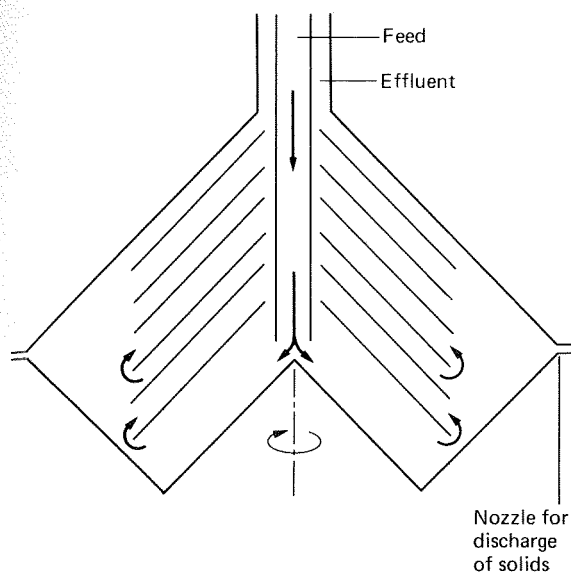


FIG. 10.19a. L.S. of disc-bowl centrifuge with nozzle discharge.

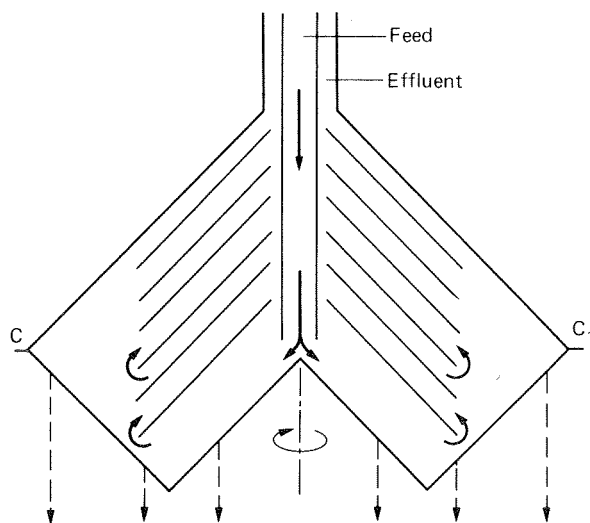


FIG. 10.19b. L.S. of disc-bowl centrifuge with intermittent discharge. (Solids discharged when rotor opens intermittently along the section C-C₁.)

mode but at a lower pressure. The degree of disruption and consequently the amount of protein released will influence the ease of subsequent separation of the product from the cell debris in high-pressure homogenizers and bead mills (Agerkvist and Enfors, 1990). A careful balance must therefore be made between percentage release of product and the difficulty and cost of further product purification.

SOLID SHEAR

Pressure extrusion of frozen micro-organisms at around -25°C through a small orifice is a well established technique at a laboratory scale using a Hughes press or an X-press to obtain small samples of enzymes or microbial cell walls. Disruption is due to a combination of liquid shear through a narrow orifice and the presence of ice crystals. Magnusson and Edebo (1976) developed a semi-continuous X-press operating with a sample temperature of -35°C and an X-press temperature of -20°C . It was possible to obtain 90% disruption with a single passage of *S. cerevisiae* using a throughput of $10\text{ kg yeast cell paste h}^{-1}$. This technique might be ideal for microbial products which are very temperature labile.

AGITATION WITH ABRASIVES

Mechanical cell disruption can also be achieved in a disintegrator containing a series of rotating discs and a charge of small beads. The beads are made of mechanically resistant materials such as glass, alumina ceramics and some titanium compounds (Fig. 10.22). In a small disintegrator, the Dyno-Muhle KD5 (Wiley A. Bachofen, Basle, Switzerland), using a flow rate of $180\text{ dm}^3\text{ h}^{-1}$, 85% disintegration of an 11% w/v suspension of *S. cerevisiae* was achieved with a single pass (Mogren *et al.*, 1974). Although temperatures of up to 35°C were recorded in the disintegrator, the specific enzyme activities were not considered to be very different from values obtained by other techniques. Dissipation of heat generated in the mill is one of the major problems in scale up, though this can generally be overcome with the provision of a cooling jacket. In another disintegrator, the Netsch LM20 mill (Netsch GmbH, Selb, Germany), the agitator blades were alternately mounted vertically and obliquely on the horizontal shaft (Fig. 10.23). A flow rate of up to $400\text{ dm}^3\text{ h}^{-1}$ was claimed for a vessel with a nominal capacity of 20 dm^3 (Rehacek and Schaefer, 1977).

FREEZING-THAWING

Freezing and thawing of a microbial cell paste will inevitably cause ice crystals to form and their expansion followed by thawing will lead to some subsequent disruption of cells. It is slow, with limited release of cellular materials, and has not often been used as a technique on its own, although it is often used in combination with other techniques. β -Glucosidase has been obtained from *S. cerevisiae* by this method (Honig and Kula, 1976). A sample of 360 g of frozen yeast

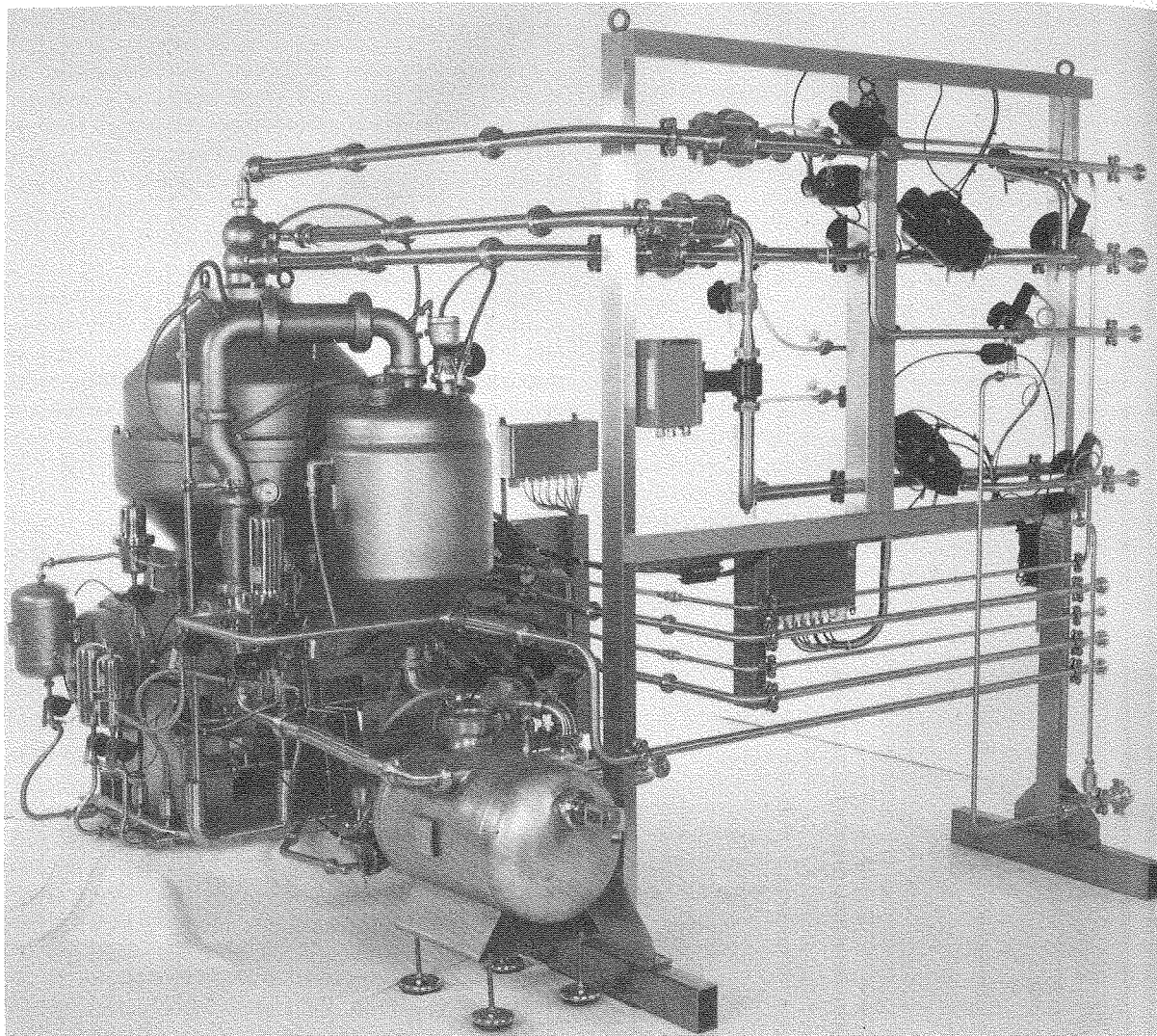


FIG. 10.20. Alfa Laval BTUX 510 disc stack centrifuge (Alfa Laval Sharples Ltd, Camberley, U.K.).

paste was thawed at 5° for 10 hours. This cycle was repeated twice before further processing.

ULTRASONICATION

High frequency vibration (~ 20 kHz) at the tip of an ultrasonication probe leads to cavitation, and shock waves thus produced cause cell disruption. The method can be very effective on a small scale, but a number of serious drawbacks make it unsuitable for large-scale operations. Power requirements are high, there is a large heating effect so cooling is needed, the probes

have a short working life and are only effective over a short range. Continuous laboratory sonicators with hold-up volumes of around 10 cm³ have been shown to be effective (James *et al.*, 1972).

Chemical methods

DETERGENTS

A number of detergents will damage the lipoproteins of the microbial cell membrane and lead to release of intracellular components. The compounds which can

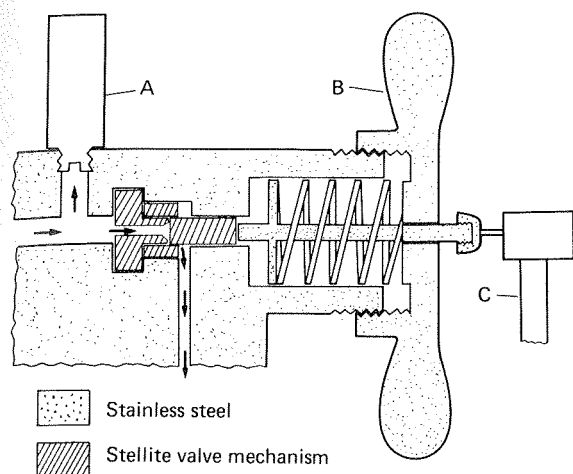


FIG. 10.21. Details of homogenizer valve assembly (Brookman, 1974). (A) 0–50,000 psi pressure transducer; (B) pressure-control handwheel; (C) linear variable displacement transformer; (→) direction of flow.

be used for this purpose include quaternary ammonium compounds, sodium lauryl sulphate, sodium dodecyl sulphate (SDS) and Triton X-100. Unfortunately, the detergents may cause some protein denaturation and may need to be removed before further purification stages can be undertaken. Pullulanase is an enzyme which is bound to the outer membrane of *Klebsiella pneumoniae*. The cells were suspended in pH 7.8 buffer and 1% sodium cholate was added. The mixture was stirred for 1 hour to solubilize most of the enzyme (Kroner *et al.*, 1978). The use of Triton X-100 in combination with guanidine-HCl is widely and effectively used for the release of cellular protein (Naglak and Wang, 1992; Hettwer and Wang, 1989), Hettwer and Wang obtaining greater than 75% protein release in less than one hour from *Escherichia coli* under fermentation conditions.

OSMOTIC SHOCK

Osmotic shock caused by a sudden change in salt concentration will cause disruption of a number of cell types. However, the effect on microbial cells is normally minimal. It has proved to be a successful technique for the extraction of luciferase from *Photobacterium fischeri* (Hastings *et al.*, 1965). A batch of 120 dm³ of broth was harvested and the cells collected as a cell paste in a Sharples centrifuge. Enzyme extraction was achieved by osmotic lysis using a ratio of 1 g of cell paste to 4 cm³ of cold distilled water with stirring for

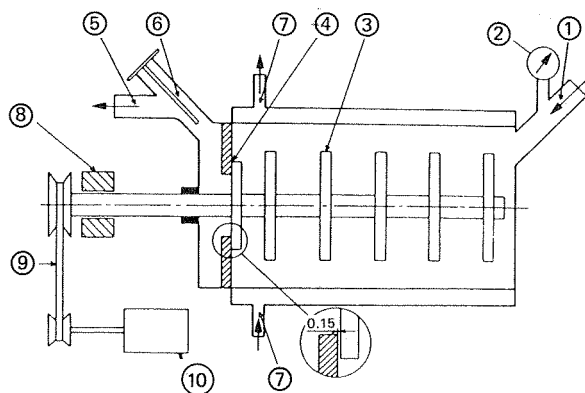


FIG. 10.22. Simplified drawing of the Dyno-Muhle KD5 (Mogren *et al.*, 1974). (1) Inlet of suspension; (2) manometer; (3) rotating disc; (4) slit for separation of glass beads from the suspension; (5) outlet of suspension; (6) thermometer; (7) cooling water, inlet and outlet; (8) bearings; (9) variable V-belt drive; (10) drive motor. Cylinder dimensions: inside length 33 cm; inside diameter 14 cm.

15 to 30 minutes. A second extraction gave a small additional yield of enzyme. Only low levels of soluble protein were released using this technique.

ALKALI TREATMENT

Alkali treatment might be used for hydrolysis of microbial cell wall material provided that the desired enzyme will tolerate a pH of 11.5 to 12.5 for 20 to 30 minutes. Darbyshire (1981) has reported the use of this technique in the extraction of L-asparaginase.

ENZYME TREATMENT

There are a number of enzymes which hydrolyse

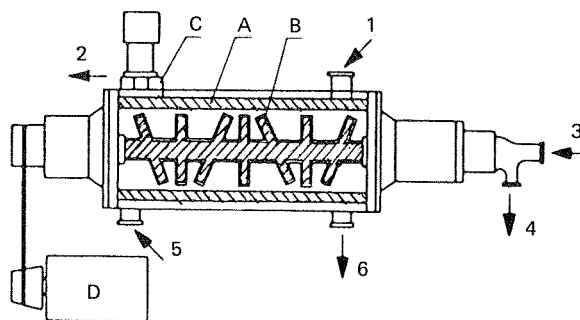


FIG. 10.23. Simplified drawing of the Netzsch model LM-20 mill (Rehacek and Schafer, 1977): A, cylindrical grinding vessel with cooling jacket; B, agitator with cooled shaft and discs; C, annular vibrating slot operator; D, variable-speed-drive motor; 1 and 2, product inlet and outlet; 3 and 4, agitator cooling inlet and outlet; 5 and 6, vessel-cooling inlet and outlet.

specific bonds in cell walls of a limited number of micro-organisms. Enzymes shown to have this activity include lysozyme and enzyme extracts from leucocytes, *Streptomyces* spp., *Micromonospora* spp. *Penicillium* spp., *Trichoderma* spp., and snails. Although this is probably one of the most gentle methods available, unfortunately it is relatively expensive and the presence of the enzyme(s) may complicate further downstream purification processes. The use of immobilized lysozyme has been investigated by a number of workers and may provide the solution to such problems (Crapisi *et al.*, 1993). Chemical and enzymic methods for the release of intracellular products have not been used widely on a large scale, with the exception of lysozyme. However, their potential for the selective release of product and that they often yield a cleaner lysate mean that they are potentially invaluable tools in the recovery of fermentation products (Andrews and Asenjo, 1987; Andrews *et al.*, 1990). Enzymes may also be used as a pretreatment to partially hydrolyse cell walls prior to cell disruption by mechanical methods.

LIQUID-LIQUID EXTRACTION

The separation of a component from a liquid mixture by treatment with a solvent in which the desired component is preferentially soluble is known as liquid-liquid extraction. The specific requirement is that a high percentage extraction of product must be obtained but concentrated in a smaller volume of solvent.

Prior to starting a large-scale extraction, it is important to find out on a small scale the solubility characteristics of the product using a wide range of solvents. A simple rule to remember is that 'like dissolves like'. The important 'likeness' as far as solubility relations are concerned is in the polarities of molecules. Polar liquids mix with each other and dissolve salts and other polar solids. The solvents for non-polar compounds are liquids of low or nil polarity.

The dielectric constant is a measure of the degree of molar polarization of a compound. If this value is known it is then possible to predict whether a compound will be polar or non-polar, with a high value indicating a highly polar compound. The dielectric constant D of a substance can be measured by determining the electrostatic capacity C of a condenser containing the substance between the plates. If C_0 is the value for the same condenser when completely evacuated then

$$D = \frac{C}{C_0}.$$

Experimentally, dielectric constants are obtained by comparing the capacity of the condenser when filled with a given liquid with the capacity of the same condenser containing a standard liquid whose dielectric constant is known very accurately. If D_1 and D_2 are the dielectric constants of the experimental and standard liquids and C_1 and C_2 are the electrostatic capacities of a condenser when filled with each of the liquids, then

$$\frac{D_1}{D_2} = \frac{C_1}{C_2}.$$

The value of D_1 can be calculated since C_1 and C_2 can be measured and D_2 is known. The dielectric constants for a number of solvents are given in Table 10.1.

The final choice of solvent will be influenced by the distribution or partition coefficient K where

$$K = \frac{\text{Concentration of solute in extract}}{\text{Concentration of solute in raffinate}}.$$

The value of K defines the ease of extraction. When there is a relatively high K value, good stability of product and good separation of the aqueous and solvent phases, then it may be possible to use a single-stage extraction system (Fig. 10.24). A value of 50 indicates that the extraction should be straightforward whereas a value of 0.1 shows that the extraction will be difficult and that a multistage process will be necessary. Unfortunately, in a number of systems the value of K is low and co-current or counter-current multistage systems

TABLE 10.1. Dielectric constants of solvents at 25°C (arranged in order of increasing polarity)

Solvent	Dielectric constant
Hexane	1.90 (least polar)
Cyclohexane	2.02
Carbon tetrachloride	2.24
Benzene	2.28
Di-ethyl ether	4.34
Chloroform	4.87
Ethyl acetate	6.02
Butan-2-ol	15.8
Butan-1-ol	17.8
Propan-1-ol	20.1
Acetone	20.7
Ethanol	24.3
Methanol	32.6
Water	78.5 (most polar)

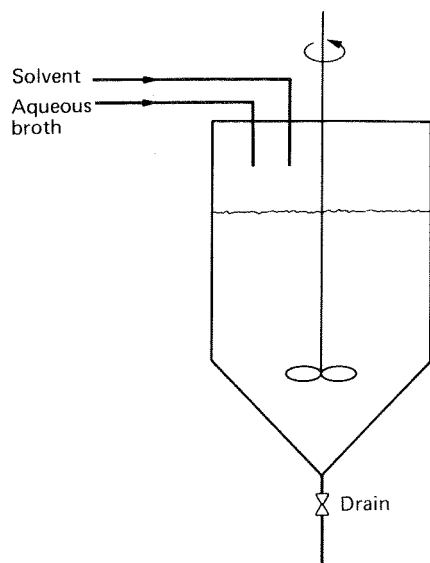


FIG. 10.24. Diagram of a single-stage extraction unit.

have to be utilized. The co-current system is illustrated in Fig. 10.25. There are n mixer/separator vessels in line and the raffinate goes from vessel 1 to vessel n . Fresh solvent is added to each stage, the feed and extracting solvent pass through the cascade in the *same* direction. Extract is recovered from each stage. Although a relatively large amount of solvent is used, a high degree of extraction is achieved.

A counter-current system is illustrated in Fig. 10.26. There are a number of mixer/separators connected in series. The extracted raffinate passes from vessel 1 to vessel n while the product-enriched solvent is flowing from vessel n to vessel 1. The feed and extracting solvent pass through the cascade in *opposite* directions. The most efficient system for solvent utilization is counter-current operation, showing a considerable advantage over batch and co-current systems. Unless there

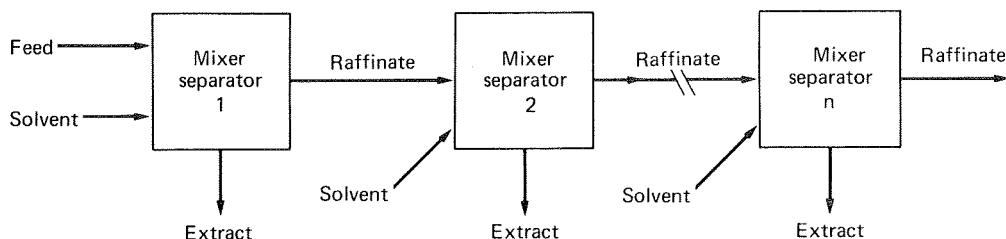
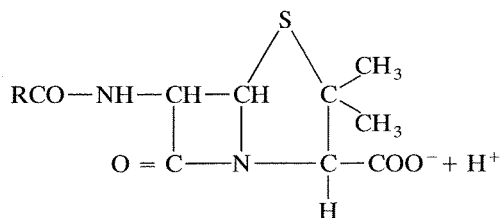


FIG. 10.25. Diagram of a co-current flow extraction system.

are special reasons the counter-current system should be used. In practice, the series of counter-current extractions are conducted in a single continuous extractor using centrifugal forces to separate the two liquid phases. The two liquid streams are forced to flow counter-current to each other through a long spiral of channels within the rotor.

The Podbielniak centrifugal extractor (Fig. 10.27) consists of a horizontal cylindrical drum revolving at up to 5000 rpm about a shaft passing through its axis. The liquids to be run counter-current are introduced into the shaft, with the heavy liquid entering the drum at the shaft while the light liquid is led by an internal route to the periphery of the drum. As the drum rotates, the heavy liquid is forced to the periphery of the drum by centrifugal action where it contacts the light liquid. The solute is transferred between the liquids and the light liquid is displaced back towards the axis of the drum. The heavy liquid is returned to the drum's axis via internal channels. The two liquid streams are then discharged via the shaft. Flow rates in excess of $100,000 \text{ dm}^3 \text{ h}^{-1}$ are possible in the largest models. Probably the most useful property of this type of extractor is the low hold-up volume of liquid in the machine compared with the throughput.

Penicillin G is an antibiotic which is recovered from fermentation broths by centrifugal counter-current solvent extraction. At neutral pHs in water penicillin is ionized:



In acid conditions this ionization is suppressed and the

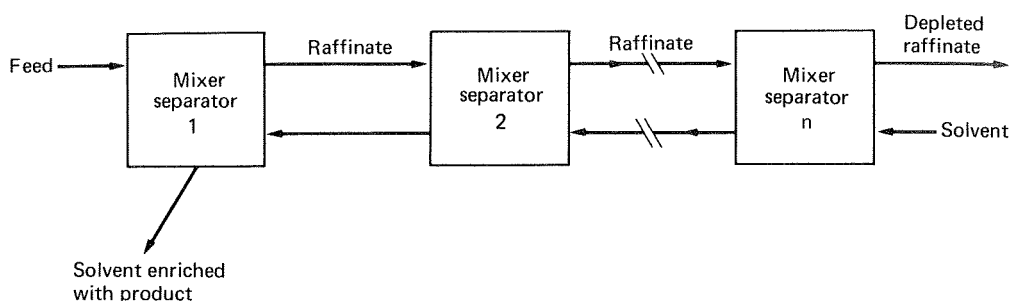


FIG. 10.26. Diagram of a counter-current extraction system.

penicillin is more soluble in organic solvents. At pH 2 to 3 the distribution ratio of total acid will be

$$K = \frac{(\text{RCOOH})_{\text{org}}}{(\text{RCOOH})_{\text{aq}} + (\text{RCOO}^-)_{\text{aq}}}$$

For penicillin this value may be as high as 40 in a suitable solvent (Podbielniak *et al.*, 1970). The penicillin extraction process may involve the four following stages:

1. Extraction of the penicillin G from the filtered broth into an organic solvent (amyl or butyl acetate or methyl iso-butyl ketone).

2. Extraction from the organic solvent into an aqueous buffer.
3. Extraction from aqueous buffer into organic solvent.
4. Extraction of the solvent to obtain the penicillin salt.

At each extraction stage progressively smaller volumes of extractant are used to achieve concentration of the penicillin (see also Fig. 10.2). Unfortunately, penicillin G has a half-life of 15 minutes at pH 2.0 at 20°. The harvested broth is therefore initially cooled to 0° to 3°. The cooled broth is then acidified to pH 2 to 3 with sulphuric or phosphoric acid immediately before extraction. This acidified broth is quickly passed through

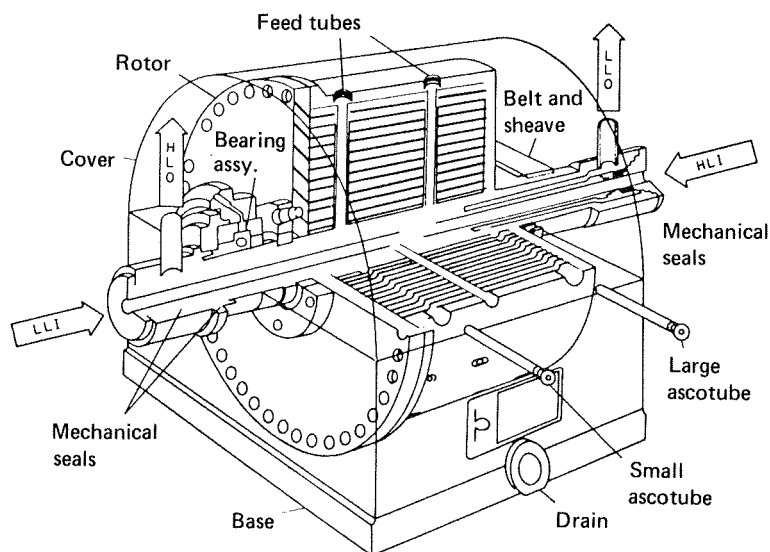
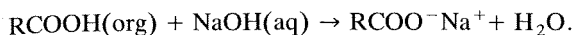


FIG. 10.27. Diagram of the Podbielniak extractor (Queener and Swartz, 1979). HLI, LLI, HLO and LLO indicate heavy and light liquid in and out.

a Podbielniak centrifugal counter-current extractor using about 20% by volume of the solvent in the counter flow. Ideally, the hold-up time should be about 60 to 90 seconds. The penicillin-rich solvent then passes through a second Podbielniak extractor counter-current to an aqueous NaOH or KOH solution (again about 20% by volume) so that the penicillin is removed to the aqueous phase (pH 7.0 to 8.0) as the salt.



These two stages may be sufficient to concentrate the penicillin adequately from a broth with a high titre. Penicillin will crystallize out of aqueous solution at a concentration of approximately 1.5×10^6 units cm^{-3} . If the broth harvested initially contains 60,000 units cm^{-3} , and two five-fold concentrations are achieved in the two extraction stages, then the penicillin liquor should crystallize. If the initial broth titre is lower than 60,000 units cm^{-3} or the extractions are not so effective, the solvent and buffer extractions will have to be repeated. At each stage the spent liquids should be checked for residual penicillin and solvent usage carefully monitored. Since the solvents are expensive and their disposal is environmentally sensitive they are recovered for recirculation through the extraction process. The success of a process may depend on efficient solvent recovery and reuse.

SOLVENT RECOVERY

A major item of equipment in an extraction process is the solvent-recovery plant which is usually a distillation unit. It is not normally essential to remove all the raffinate from the solvent as this will be recycled through the system. In some processes the more difficult problem will be to remove all the solvent from the raffinate because of the value of the solvent and problems which might arise from contamination of the product.

Distillation may be achieved in three stages:

1. Evaporation, the removal of solvent as a vapour from a solution.
2. Vapour-liquid separation in a column, to separate the lower boiling more volatile component from other less volatile components.
3. Condensation of the vapour, to recover the more volatile solvent fraction.

Evaporation is the removal of solvent from a solution by the application of heat to the solution. A wide

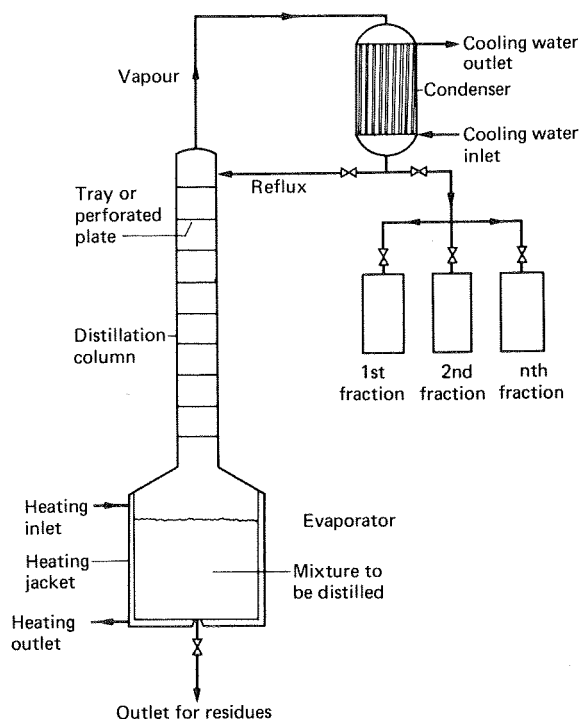


FIG. 10.28. Diagram of a batch distillation plant with a tray or perforated-plate column.

range of evaporators is available. Some are operated on a batch basis and others continuously. Most industrial evaporators employ tubular heating surfaces. Circulation of the liquid past the heating surfaces may be induced by boiling or by mechanical agitation. In batch distillation (Fig. 10.28) the vapour from the boiler passes up the column and is condensed. Part of the condensate will be returned as the reflux for counter-current contact with the rising vapour in the column. The distillation is continued until a satisfactory recovery of the lower-boiling (more volatile) component(s) has been accomplished. The ratio of condensate returned to the column as reflux to that withdrawn as product is, along with the number of plates or stages in the column, the major method of controlling the product purity. A continuous distillation (Fig. 10.29) is initially begun in a similar way as with a batch distillation, but no condensate is withdrawn initially. There is total reflux of the condensate until ideal operating conditions have been established throughout the column. At this stage the liquid feed is fed into the column at an intermediate level. The more volatile components move upwards as vapour and are condensed, followed by partial reflux of

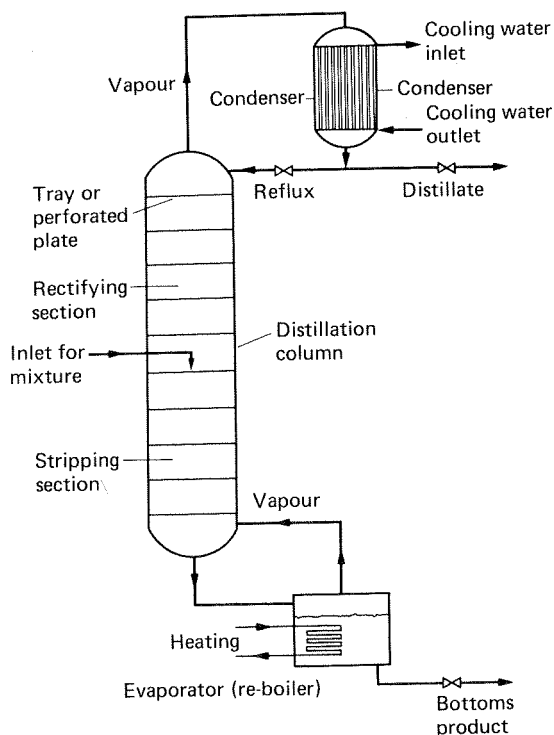


FIG. 10.29. Diagram of a continuous distillation plant with a tray or perforated-plate column.

the condensate. Meanwhile, the less volatile fractions move down the column to the evaporator (reboiler). At this stage part of the bottoms fraction is continuously withdrawn and part is reboiled and returned to the column.

Counter-current contacting of the vapour and liquid streams is achieved by causing:

- vapour to be dispersed in the liquid phase (plate or tray column),
- liquid to be dispersed in a continuous vapour phase (packed column).

The plate or tray column consists of a number of distinct chambers separated by perforated plates or trays. The rising vapour bubbles through the liquid which is flowing across each plate, and is dispersed into the liquid from perforations (sieve plates) or bubble caps. The liquid flows across the plates and reaches the reboiler by a series of overflow weirs and down pipes.

A packed tower is filled with a randomly packed material such as rings, saddles, helices, spheres or beads. Their dimensions are approximately one-tenth

to one-fiftieth of the diameter of the column and are designed to provide a large surface area for liquid-vapour contacting and high voidage to allow high throughput of liquid and vapour.

The heat input to a distillation column can be considerable. The simplest ways of conserving heat are to preheat the initial feed by a heat exchanger using heat from:

- the hot vapours at the top of the column,
- heat from the bottoms fraction when it is being removed in a continuous process,
- a combination of both.

Since it is beyond the scope of this text to consider the distillation process more fully the reader is therefore directed to Coulson and Richardson (1991).

TWO-PHASE AQUEOUS EXTRACTION

Liquid-liquid extraction is a well established technology in chemical processing and in certain sectors of biochemical processing. However, the use of organic solvents has limited application in the processing of sensitive biologicals. Aqueous two-phase systems, on the other hand, have a high water content and low interfacial surface tension and are regarded as being biocompatible (Mattiasson and Ling, 1987).

Two-phase aqueous systems have been known since the late nineteenth century, and a large variety of natural and synthetic hydrophilic polymers are used today to create two (or more) aqueous phases. Phase separation occurs when hydrophilic polymers are added to an aqueous solution, and when the concentrations exceed a certain value two immiscible aqueous phases are formed. Settling time for the two phases can be prolonged, depending on the components used and vessel geometry. Phase separation can be improved by using centrifugal separators (Huddleston *et al.*, 1991), or novel techniques such as magnetic separators (Wikstrom *et al.*, 1987).

Many systems are available:

- Non-ionic polymer/non-ionic polymer/water, e.g. polyethylene glycol/dextran.
- Polyelectrolyte/non-ionic polymer/water, e.g. sodium carboxymethyl cellulose/polyethylene glycol.
- Polyelectrolyte/polyelectrolyte/water, e.g. sodium dextran sulphate/sodium carboxymethyl cellulose

- (iv) Polymer/low molecular weight component/water, e.g. dextran/propyl alcohol.

The distribution of a solute species between the phases is characterized by the partition coefficient, and is influenced by a number of factors such as temperature, polymer (type and molecular weight), salt concentration, ionic strength, pH and properties (e.g. molecular weight) of the solute. As the goal of any extraction process is to selectively recover and concentrate a solute, affinity techniques such as those applied in chromatographic processes can be used to improve selectivity. Examples include the use of PEG-NADH derivatives in the extraction of dehydrogenases, *p*-aminobenzamidine in the extraction of trypsin and cibacron blue in the extraction of phosphofructokinase. It is possible to use different ligands in the two phases leading to an increase in selectivity or the simultaneous recovery and separation of several species (Cabral and Aires-Barros, 1993).

Two phase aqueous systems have found application in the purification of many solutes; proteins, enzymes (Gonzalez *et al.*, 1990; Guan *et al.*, 1992), cells and subcellular particles, and in extractive bioconversions. Several aqueous two-phase systems for handling large-scale protein separation have emerged, the majority of which use PEG as the upper phase forming polymer with either dextran, concentrated salt solution or hydroxypropyl starch as the lower phase forming material (Mattiasson and Kaul, 1986). Hustedt *et al.* (1988) demonstrated the application of continuous cross-current extraction of enzymes (fumarase and penicillin acylase) by aqueous two-phase systems at production scale.

SUPERCritical FLUID EXTRACTION

The technique of supercritical fluid extraction utilizes the dissolution power of supercritical fluids, i.e. fluids above their critical temperature and pressure. Its advantages include the use of moderate temperatures, and that several cheap and non-toxic fluids are available.

Supercritical fluids are used in the extraction of hop oils, caffeine, vanilla, vegetable oils and β -carotene. It has also been shown experimentally that the extraction of certain steroids and chemotherapeutic drugs can be achieved using supercritical fluids. Other current and potential uses include the removal of undesirable substances such as pesticide residues, removal of bacteriostatic agents from fermentation broths, the recovery

of organic solvents from aqueous solutions, cell disintegration, destruction and treatment of industrial wastes and liposome preparation. There are, however, a number of significant disadvantages in the utilization of this technology:

- (i) Phase equilibria of the solvent/solute system is complex, making design of extraction conditions difficult.
- (ii) The most popular solvent (carbon dioxide) is non-polar and is therefore most useful in the extraction of non-polar solutes. Though co-solvents can be added for the extraction of polar compounds, they will complicate further downstream processing.
- (iii) The use of high pressures leads to capital costs for plant, and operating costs may also be high.

Thus, the number of commercial processes utilizing supercritical fluid extraction is relatively small, due mainly to the existence of more economical processes. However, its use is likely to increase in some sectors, for example the recovery of high value biologicals, when conventional extractions are inappropriate, and in the treatment of toxic wastes (Bruno *et al.*, 1993).

CHROMATOGRAPHY

In many fermentation processes, chromatographic techniques are used to isolate and purify relatively low concentrations of metabolic products. In this context, chromatography will be concerned with the passage and separation of different solutes as liquid is passed through a column, i.e. *liquid chromatography*. Depending on the mechanism by which the solutes may be differentially held in a column, the techniques can be grouped as follows:

- (a) Adsorption chromatography.
- (b) Ion-exchange chromatography.
- (c) Gel permeation chromatography.
- (d) Affinity chromatography.
- (e) Reverse phase chromatography.
- (f) High performance liquid chromatography.

Chromatographic techniques are also used in the final stages of purification of a number of products. The scale-up of chromatographic processes can prove difficult, and there is much current interest in the use of mathematical models and computer programmes to

translate data obtained from small-scale processes into operating conditions for larger scale applications (Cowan *et al.*, 1986, 1987).

Adsorption chromatography

Adsorption chromatography involves binding of the solute to the solid phase primarily by weak Van der Waals forces. The materials used for this purpose to pack columns include inorganic adsorbents (active carbon, aluminium oxide, aluminium hydroxide, magnesium oxide, silica gel) and organic macro-porous resins. Adsorption and affinity chromatography are mechanistically identical, but are strategically different. In affinity systems selectivity is designed rationally whilst in adsorption selectivity must be determined empirically.

Di-hydro-streptomycin can be extracted from filtrates using activated charcoal columns. It is then eluted with methanolic hydrochloric acid and purified in further stages (Nakazawa *et al.*, 1960). Some other applications for small-scale antibiotic purification are quoted by Weinstein and Wagman (1978). Active carbon may be used to remove pigments to clarify broths. Penicillin-containing solvents may be treated with 0.25 to 0.5% active carbon to remove pigments and other impurities (Sylvester and Coghill, 1954).

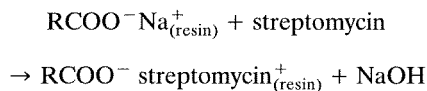
Macro-porous adsorbents have also been tested. The first synthetic organic macro-porous adsorbents, the Amberlite XAD resins, were produced by Rohm and Haas in 1965. These resins have surface polarities which vary from non-polar to highly polar and do not possess any ionic functional groups. Voser (1982) considers their most interesting application to be in the isolation of hydrophilic fermentation products. He stated that these resins would be used at Ciba-Geigy in recovery of cephalosporin C (acidic amino acid), cefotiam (basic amino acid), desferrioxamine B (basic hydroxamic acid) and paramethasone (neutral steroid).

Ion exchange

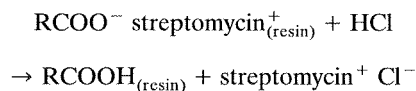
Ion exchange can be defined as the reversible exchange of ions between a liquid phase and a solid phase (ion-exchange resin) which is not accompanied by any radical change in the solid structure. Cationic ion-exchange resins normally contain a sulphonic acid, carboxylic acid or phosphonic acid active group. Carboxy-methyl cellulose is a common cation exchange resin. Positively charged solutes (e.g. certain proteins) will bind to the resin, the strength of attachment de-

pending on the net charge of the solute at the pH of the column feed. After deposition solutes are sequentially washed off by the passage of buffers of increasing ionic strength or pH. Anionic ion-exchange resins normally contain a secondary amine, quaternary amine or quaternary ammonium active group. A common anion exchange resin, DEAE (diethylaminoethyl) cellulose is used in a similar manner to that described above for the separation of negatively charged solutes. Other functional groups may also be attached to the resin skeleton to provide more selective behaviour similar to that of affinity chromatography. The appropriate resin for a particular purpose will depend on various factors such as bead size, pore size, diffusion rate, resin capacity, range of reactive groups and the life of the resin before replacement is necessary. Weak-acid cation ion-exchange resins can be used in the isolation and purification of streptomycin, neomycin and similar antibiotics.

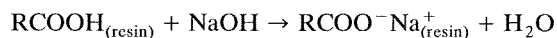
In the recovery of streptomycin, the harvested filtrate is fed on to a column of a weak-acid cationic resin such as Amberlite IRC 50 which is in the sodium form. The streptomycin is adsorbed on to the column and the sodium ions are displaced.



Flow rates of between 10 and 30 bed volumes per hour have been used. The resin bed is now rinsed with water and eluted with dilute hydrochloric acid to release the bound streptomycin.



A slow flow is used to ensure the highest recovery of streptomycin using the smallest volume of eluant. In one step the antibiotic has been both purified and concentrated, maybe more than 100-fold. The resin column is regenerated to the sodium form by passing an adequate volume of NaOH slowly through the column and rinsing with distilled water to remove excess sodium ions.



The resin can have a capacity of 1 g of streptomycin g⁻¹ resin. Commercially, it is not economic to regenerate the resin completely, therefore the capacity will be reduced. In practice, the filtered broth is taken through two columns in series while a third is being eluted and

regenerated. When the first column is saturated, it is isolated for elution and regeneration while the third column is brought into operation.

Details for isolation of some other antibiotics are given in Weinstein and Wagman (1978). Ion-exchange chromatography may be combined with HPLC in, for example, the purification of somatotropin using DEAE cellulose columns and β -urogastrone in multi-gram quantities using a cation exchange column (Brewer and Larsen, 1987).

Gel permeation

This technique is also known as gel exclusion and gel filtration. Gel permeation separates molecules on the basis of their size. The smaller molecules diffuse into the gel more rapidly than the larger ones, and penetrate the pores of the gel to a greater degree. This means that once elution is started, the larger molecules which are still in the voids in the gel will be eluted first. A wide range of gels are available, including cross-linked dextrans (Sephadex and Sephacryl) and cross-linked agarose (Sephacrose) with various pore sizes depending on the fractionation range required.

One early industrial application, although on a relatively small scale, was the purification of vaccines (Latham *et al.*, 1967). Tetanus and diphtheria broths for batches of up to 100,000 human doses are passed through a 13 dm³ column of G 100 followed by a 13 dm³ column of G 200. This technique yields a fairly pure fraction which is then concentrated ten-fold by pressure dialysis to remove the eluant buffer (Na₂HPO₄).

Affinity chromatography

Affinity chromatography is a separation technique with many applications since it is possible to use it for separation and purification of most biological molecules on the basis of their function or chemical structure. This technique depends on the highly specific interactions between pairs of biological materials such as enzyme-substrate, enzyme-inhibitor, antigen-antibody, etc. The molecule to be purified is specifically adsorbed from, for example, a cell lysate applied to the affinity column by a binding substance (ligand) which is immobilized on an insoluble support (matrix). Eluent is then passed through the column to release the highly purified and concentrated molecule. The ligand is at-

tached to the matrix by physical absorption or chemically by a covalent bond. The pore size and ligand location must be carefully matched to the size of the product for effective separation. The latter method is preferred whenever possible. Porath (1974) and Yang and Tsao (1982) have reviewed methods and coupling procedures.

Coupling procedures have been developed using cyanogen bromide, bisoxiranes, disaziridines and periodates, for matrixes of gels and beads. Four polymers which are often used for matrix materials are agarose, cellulose, dextrose and polyacrylamide. Agarose activated with cyanogen bromide is one of the most commonly used supports for the coupling of amino ligands. Silica based solid phases have been shown to be an effective alternative to gel supports in affinity chromatography (Mohan and Lyddiatt, 1992).

Purification may be several thousand-fold with good recovery of active material. The method can however be quite costly and time consuming, and alternative affinity methods such as affinity cross-flow filtration, affinity precipitation and affinity partitioning may offer some advantages (Janson, 1984; Luong *et al.*, 1987). Affinity chromatography was used initially in protein isolation and purification, particularly enzymes. Since then many other large-scale applications have been developed for enzyme inhibitors, antibodies, interferon and recombinant proteins (Janson and Hedman, 1982; Ostlund, 1986; Folea-Wasserman *et al.*, 1987; Nachman *et al.*, 1992), and on a smaller scale for nucleic acids, cell organelles and whole cells (Yang and Tsao, 1982). In the scale-up of affinity chromatographic processes (Katoh, 1987) bed height limits the superficial velocity of the liquid, thus scale-up requires an increase in bed diameter or adsorption capacity.

Reverse phase chromatography (RPC)

This chromatographic method utilizes a solid phase (e.g. silica) which is modified so as to replace hydrophilic groups with hydrophobic alkyl chains. This allows the separation of proteins according to their hydrophobicity. More-hydrophobic proteins bind most strongly to the stationary phase and are therefore eluted later than less-hydrophobic proteins. The alkyl groupings are normally eight or eighteen carbons in length (C₈ and C₁₈). RPC can also be combined with affinity techniques in the separation of, for example, proteins and peptides (Davankov *et al.*, 1990).

High performance liquid chromatography (HPLC)

HPLC is a high resolution column chromatographic technique. Improvements in the nature of column packing materials for a range of chromatographic techniques (e.g. gel permeation and ion-exchange) yield smaller, more rigid and more uniform beads. This allows packing in columns with minimum spaces between the beads, thus minimizing peak broadening of eluted species. It was originally known as high *pressure* liquid chromatography because of the high pressures required to drive solvents through silica based packed beds. Improvements in performance led to the name change and its widespread use in the separation and purification of a wide range of solute species, including bio-molecules. HPLC is distinguished from liquid chromatography by the use of improved media (in terms of their selectivity and physical properties) for the solid (stationary) phase through which the mobile (fluid) phase passes.

The stationary phase must have high surface area/unit volume, even size and shape and be resistant to mechanical and chemical damage. However, it is factors such as these which lead to high pressure requirements and cost. This may be acceptable for analytical work, but not for preparative separations. Thus, in preparative HPLC some resolution is often sacrificed (by the use of larger stationary-phase particles) to reduce operating and capital costs. For very high value products large-scale HPLC columns containing analytical media have been used.

Affinity techniques can be merged with HPLC to combine the selectivity of the former with the speed and resolving power of the latter (Forstecher *et al.*, 1986; Shojaosadaty and Lyddiatt, 1987).

Continuous chromatography

Although the concept of continuous enzyme isolation is well established (Dunnill and Lilly, 1972), the stage of least development is continuous chromatography. Fox *et al.* (1969) developed a continuous-fed column for this purpose (Fig. 10.30). It consisted of two concentric cylindrical sections clamped to a base plate. The space (1 cm wide) between the two sections was packed with the appropriate resin or gel giving a total column capacity of 2.58 dm³. A series of orifices in the circumference of the base plate below the column space led to collecting vessels. The column assembly was rotated in a slow-moving turntable (0.4–2.0 rpm). The mixture for separation was fed to the apparatus by

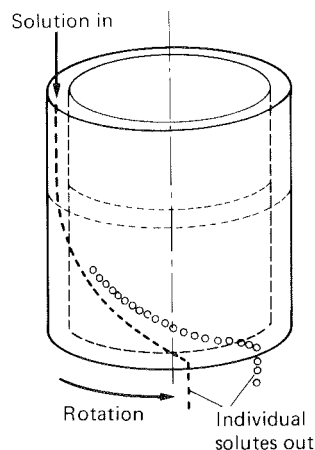


FIG. 10.30. The principle of continuous-partition chromatography. ---, faster-moving component; ○ ○, slower-moving component (Fox, 1969).

an applicator rotating at the same speed as the column, thus allowing application at a fixed point, while the eluent was fed evenly to the whole circumference of the column. The components of a mixture separated as a series of helical pathways, which varied with the retention properties of the constituent components. This method gave a satisfactory separation and recovery but the consumption of eluent and the unreliable throughput rate were not considered to be satisfactory for a large-scale method (Nicholas and Fox, 1969; Dunnill and Lilly, 1972). However, the development of such continuous separation equipment suitable for large-scale extraction would considerably simplify the use of chromatographic separation.

MEMBRANE PROCESSES

Ultrafiltration and reverse osmosis

Both processes utilize semi-permeable membranes to separate molecules of different sizes and therefore act in a similar manner to conventional filters.

Ultrafiltration

Ultrafiltration can be described as a process in which solutes of high molecular weight are retained when the solvent and low molecular weight solutes are forced under hydraulic pressure (around 7 atmospheres) through a membrane of a very fine pore size. It is

therefore used for product concentration and purification. A range of membranes made from a variety of polymeric materials, with different molecular weight cut-offs (500 to 500,000), are available which makes possible the separation of macro-molecules such as proteins, enzymes, hormones and viruses. It is practical only to separate molecules whose molecular weights are a factor of ten different due to variability in pore size (Heath and Belfort, 1992). Because the flux through such a membrane is inversely proportional to its thickness, asymmetric membranes are used where the membrane ($\sim 0.3 \mu\text{m}$ thick) is supported by a mesh around 0.3 mm thick.

When considering the feasibility of ultrafiltration it is important to remember that factors other than the molecular weight of the solute affect the passage of molecules through the membranes (Melling and Westmacott, 1972). There may be concentration polarization caused by accumulation of solute at the membrane surface which can be reduced by increasing the shear forces at the membrane surface either by conventional agitation or by the use of a cross-flow system (see previous section). Secondly a slurry of protein may accumulate on the membrane surface forming a gel layer which is not easily removed by agitation. Formation of the gel layer may be partially controlled by careful choice of conditions such as pH (Bailey and Ollis, 1986). Finally, equipment and energy costs may be considerable because of the high pressures necessary; this also limits the life of ultrafiltration membranes.

There are numerous examples of the use of ultrafiltration for the recovery of bio-molecules: viruses (Weiss, 1980), enzymes (Atkinson and Mavituna, 1991), antibiotics (Pandey *et al.*, 1985). Details of large scale applications are given by Lacey and Loeb (1972) and by Ricketts *et al.* (1985). Affinity ultrafiltration (Luong *et al.*, 1987; Luong and Nguyen, 1992) is a novel separation process developed to circumvent difficulties in affinity chromatography. It offers high selectivity, yield and concentration, but it is an expensive batch process and scale up is difficult.

Reverse osmosis

Reverse osmosis is a separation process where the solvent molecules are forced by an applied pressure to flow through a semi-permeable membrane in the opposite direction to that dictated by osmotic forces, and hence is termed reverse osmosis. It is used for the concentration of smaller molecules than is possible by

ultrafiltration. Concentration polarization is again a problem and must be controlled by increased turbulence at the membrane surface.

Liquid membranes

Liquid membranes are insoluble liquids (e.g. an organic solvent) which are selective for a given solute and separate two other liquid phases. Extraction takes place by the transport of solute from one liquid to the other. They are of great interest in the extraction and purification of biologicals for the following reasons:

- (a) Large area for extraction.
- (b) Separation and concentration are achieved in one step.
- (c) Scale-up is relatively easy.

Their use has been reported in the extraction of lactic acid (Chaudhuri and Pyle, 1990) and citric acid using a supported liquid membrane (Sirman *et al.*, 1990). The utilization of selective carriers to transport specific components across the liquid membrane at relatively high rates has increased interest in recent years (Strathmann, 1991). Liquid membranes may also be used in cell and enzyme immobilization, and thus provide the opportunity for combined production and isolation/extraction in a single unit (Mohan and Li, 1974, 1975). The potential use of liquid membranes has also been described for the production of alcohol reduced beer as having little effect on flavour or the physico-chemical properties of the product (Etuk and Murray, 1990).

DRYING

The drying of any product (including biological products) is often the last stage of a manufacturing process (McCabe *et al.*, 1984; Coulson and Richardson, 1991). It involves the final removal of water from a heat-sensitive material ensuring that there is minimum loss in viability, activity or nutritional value. Drying is undertaken because:

- (a) The cost of transport can be reduced.
- (b) The material is easier to handle and package.
- (c) The material can be stored more conveniently in the dry state.

A detailed review of the theory and practice of drying can be found in Perry and Green (1984). It is important that as much water as possible is removed initially by centrifugation or in a filter press to minimize heating costs in the drying process. Driers can be classified by the method of heat transfer to the product and the degree of agitation of the product. In contact driers the product is contacted with a heated surface. An example of this type is the drum drier (Fig. 10.31), which may be used for more temperature stable bio-products. A slurry is run onto a slowly rotating steam heated drum, evaporation takes place and the dry product is removed by a scraper blade in a similar manner as for rotary vacuum filtration. The solid is in contact with the heating surface for 6–15 seconds and heat transfer coefficients are generally between 1 and 2 kW m⁻² K⁻¹. Vacuum drum driers can be used to lower the temperature of drying.

A spray drier (Fig. 10.32) is most widely used for drying of biological materials when the starting material is in the form of a liquid or paste. The material to be dried does not come into contact with the heating surfaces, instead, it is atomized into small droplets through for example a nozzle or by contact with a rotating disc. The wide range of atomizers available is described in Coulson and Richardson (1991). The droplets then fall into a spiral stream of hot gas at 150° to 250°. The high surface area:volume ratio of the droplets results in a rapid rate of evaporation and complete drying in a few seconds, with drying rate and product size being directly related to droplet size produced by the atomizer. The evaporative cooling effect prevents the material from becoming overheated and damaged. The gas-flow rate must be carefully regulated

so that the gas has the capacity to contain the required moisture content at the cool-air exhaust temperature (75° to 100°). In most processes the recovery of very small particles from the exit gas must be conducted using cyclones or filters. This is especially important for containment of biologically active compounds. The jet spray drier is particularly suited to handling heat sensitive materials. Operating at a temperature of around 350°, residence times are approximately 0.01 seconds because of the very fine droplets produced in the atomizing nozzle.

Spray driers are the most economical available for handling large volumes, and it is only at feed rates below 6 kg min⁻¹ that drum driers become more economic.

Freeze drying is an important operation in the production of many biologicals and pharmaceuticals. The material is first frozen and then dried by sublimation in a high vacuum. The great benefit of this technique is that it does not harm heat sensitive materials. The process is often termed lyophilization when the solvent being evaporated is water.

Fluidized bed driers are used increasingly in the

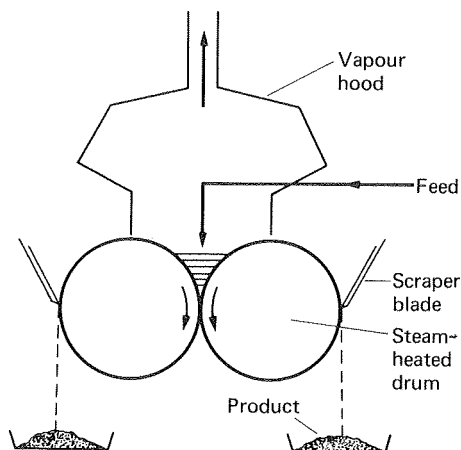


FIG. 10.31. Cross-section of a drum drier.

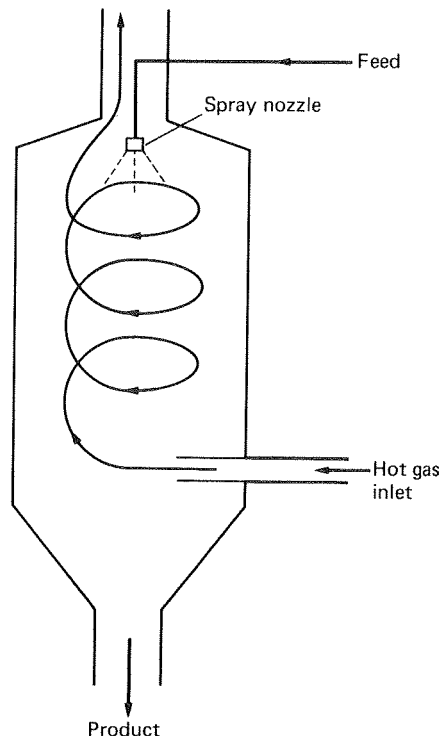


FIG. 10.32. Counter-current spray drier.

pharmaceutical industry. Heated air is fed into a chamber of fluidized solids, to which wet material is continuously added and dry material continuously removed. Very high mass-transfer rates are achieved, giving rapid evaporation and allowing the whole bed to be maintained in a dry condition.

CRYSTALLIZATION

Crystallization is an established method used in the initial recovery of organic acids and amino acids, and more widely used for final purification of a diverse range of compounds.

In citric acid production, the filtered broth is treated with $\text{Ca}(\text{OH})_2$ so that the relatively insoluble calcium citrate crystals will be precipitated from solution. Checks are made to ensure that the $\text{Ca}(\text{OH})_2$ has a low magnesium content, since magnesium citrate is more soluble and would remain in solution. The calcium citrate is filtered off and treated with sulphuric acid to precipitate the calcium as the insoluble sulphate and release the citric acid. After clarification with active carbon, the aqueous citric acid is evaporated to the point of crystallization (Lockwood and Irwin, 1964; Sodeck *et al.*, 1981; Atkinson and Mavituna, 1991). Crystallization is also used in the recovery of amino acids; Samejima (1972) has reviewed methods for glutamic acid, lysine and other amino acids. The recovery of cephalosporin C as its sodium or potassium salt by crystallization has been described by Wildfeuer (1985).

WHOLE BROTH PROCESSING

The concept of recovering a metabolite directly from an unfiltered fermentation broth is of considerable interest because of its simplicity, the reduction in process stages and the potential cost savings. It may also be possible to remove the desired fermentation product continuously from a broth during fermentation so that inhibitory effects due to product formation and product degradation can be minimized throughout the production phase (Roffler *et al.*, 1984; Diaz, 1988).

Bartels *et al.* (1958) developed a process for adsorption of streptomycin on to a series of cationic ion-exchange resin columns directly from the fermentation broth, which had only been screened to remove large particles so that the columns would not become blocked. This procedure could only be used as a batch process. Belter *et al.* (1973) developed a similar process

for the recovery of novobiocin. The harvested broth was first filtered through a vibrating screen to remove large particles. The broth was then fed into a continuous series of well-mixed resin columns fitted with screens to retain the resin particles, plus the absorbed novobiocin, but allow the streptomycete filaments plus other small particulate matter to pass through. The first resin column was removed from the extraction line after a predetermined time and eluted with methanolic ammonium chloride to recover the novobiocin.

Karr *et al.* (1980) developed a reciprocating plate extraction column (Fig. 10.33) to use for whole broth processing of a broth containing 1.4 g dm^{-3} of a slightly soluble organic compound and 4% undissolved solids provided that chloroform or methylene chloride were used for extraction. Methyl-iso-butyl ketone, diethyl ketone and iso-propyl acetate were shown to be more efficient solvents than chloroform for extracting the active compound, but they presented problems since they also extracted impurities from the mycelia, making it necessary to filter the broth before beginning the solvent extraction. Considerable economies were claimed in a comparison with a process using a Podbielniak extractor, in investment, maintenance costs, solvent usage and power costs but there was no significant difference in operating labour costs.

An alternative approach is to remove the metabolite continuously from the broth during the fermentation. Cycloheximide production by *Streptomyces griseus* has been shown to be affected by its own feed-back regulation (Kominek, 1975). Wang *et al.* (1981) have tested two techniques at laboratory scale for improving production of cycloheximide. In a dialysis method (Fig. 10.34), methylene chloride was circulated in a dialysis tubing loop which passed through a 10 dm^{-3} fermenter. Cycloheximide in the fermentation broth was extracted into the methylene chloride. It was shown that the product yield could be almost doubled by this dialysis-solvent extraction method to over $1200 \mu\text{g cm}^{-3}$ as compared with a control yield of approximately $700 \mu\text{g cm}^{-3}$. In a resin method, sterile beads of XAD-7, an acrylic resin, as dispersed beads or beads wrapped in an ultrafiltration membrane, were put in fermenters 48 hours after inoculation. Some of the cycloheximide formed in the broth is absorbed by the resin. Recovery of the antibiotic from the resin is achieved by solvents or by changing the temperature or pH. When assayed after harvesting, the control (without resin) had a bioactivity of $750 \mu\text{g cm}^{-3}$. Readings of total bioactivity (from beads and broth) for the bead treatment and the membrane-wrapped bead treatments

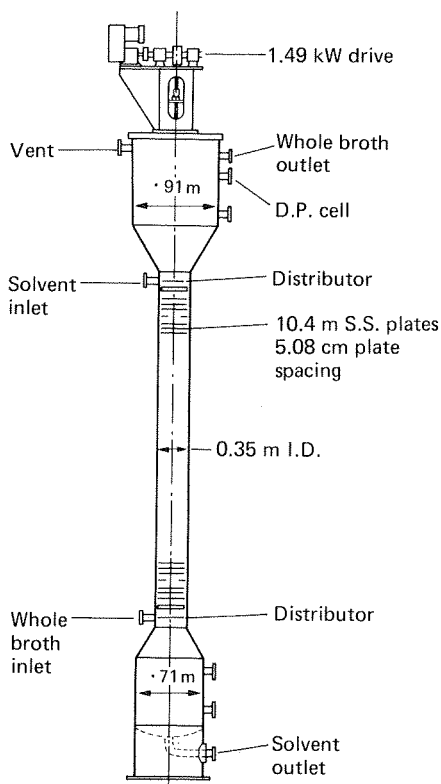


FIG. 10.33a. Diagram of a 0.35-m internal diameter reciprocating plate column (Karr *et al.*, 1980).

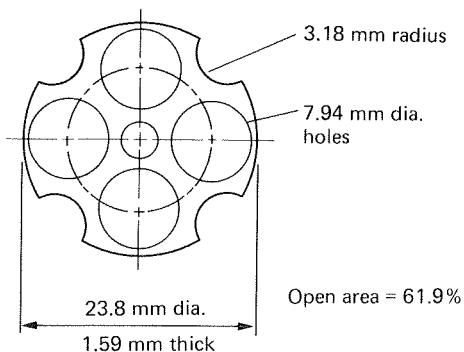


FIG. 10.33b. Plan of a 23.8-m stainless-steel plate for a 25-mm diameter reciprocating plate test column (Karr *et al.*, 1980).

were $1420 \mu\text{g cm}^{-3}$ and $1790 \mu\text{g cm}^{-3}$ respectively.

Roffler *et al.* (1984) reviewed the use of a number of techniques for the *in-situ* recovery of fermentation products:

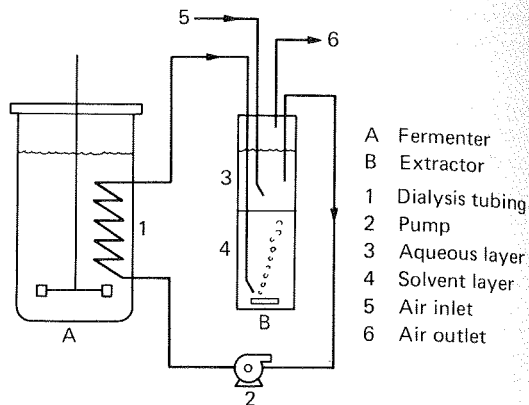


FIG. 10.34. Dialysis-extraction fermentation system (Wang *et al.*, 1981).

- Vacuum and flash fermentations for the direct recovery of ethanol from fermentation broths.
- Extractive fermentation (liquid-liquid and two-phase aqueous) for the recovery of ethanol, organic acids and toxin produced by *Clostridium tetani*.
- Adsorption for the recovery of ethanol and cycloheximide.
- Ion-exchange in the extraction of salicylic acid and antibiotics.
- Dialysis fermentation in the selective recovery of lactic acid, salicylic acid and cycloheximide.

Hansson *et al.* (1994) have used an expanded adsorption bed for the recovery of a recombinant protein produced by *E. coli* directly from the fermentation broth. The protein was produced in high yields (550 mg dm^{-3}) and $> 90\%$ recovery together with concentration (volume reduction) and removal of cells was achieved on the expanded bed. Affinity chromatography was used for further purification, and again an overall yield of $> 90\%$ obtained.

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Effluent Treatment

INTRODUCTION

EVERY fermentation plant utilizes raw materials which are converted to a variety of products. Depending on the individual process, varying amounts of a range of waste materials are produced. Typical wastes might include unconsumed inorganic and organic media components, microbial cells and other suspended solids, filter aids, waste wash water from cleansing operations, cooling water, water containing traces of solvents, acids, alkalis, human sewage, etc. Historically, it was possible to dispose of wastes directly to a convenient area of land or into a nearby watercourse. This cheap and simple method of disposal is now very rarely possible, nor is it environmentally desirable. With increasing density of population and industrial expansion, together with greater awareness of the damage caused by pollution, the need for treatment and controlled disposal of waste has, and will, continue to grow. Water authorities and similar bodies have become more active in combating pollution caused by domestic and industrial wastes. Legislation in all developed countries now regulates the discharge of wastes, be they gas, liquid or solid (Fisher, 1977; Hill, 1980; Masters, 1991; Brown, 1992). In the U.K., much of the legislation pertaining to waste disposal and pollution is embraced by the Environmental Pollution Act 1990 (HMSO, 1990, 1991). Further information on legislation and environmental policy can be found in the following texts; Haigh (1990), Tromans (1991) and Hughes (1992).

With liquid wastes, it may be possible to dispose of untreated effluents to a municipal sewage treatment works (STW). Obviously, much will depend on the composition, strength and volumetric flow rate of the effluent. STWs are planned to operate with an effluent

of a reasonably constant composition at a steady flow rate. Thus, if the discharge from an industrial process is large in volume and intermittently produced it may be necessary to install storage tanks on site to regulate the effluent flow. In some locations, municipal sewers are not available or the effluent may be of such a composition that the wastewater treatment company or regulatory authority requires some form of pretreatment before discharge to its sewers. In these cases an effluent-treatment plant will have to be installed at the factory. Whatever the pollutant load of the liquid effluent, its discharge to a sewer will be a cost centred activity, and will incur charges from the treatment company.

Normally, fermentation effluents do not contain toxic materials which directly affect the aquatic flora or fauna. Unfortunately, most of the effluents do contain high levels of organic matter which are readily oxidized by microbial attack and so drastically deplete the dissolved oxygen concentration in the receiving water unless there is a large dilution factor. This can be shown by the oxygen sag curve in Fig. 11.1. Different aquatic species have varying tolerances to depleted oxygen levels, and as a consequence some species will die off in specific stretches of the receiving water, and in other regions a different population capable of growth at lower oxygen levels will develop.

Effluents may be treated in a variety of ways, as will be outlined later in this chapter. In a number of processes it may be possible to recover waste organic material as a solid and sell it as a by-product which may be an animal feed supplement or a nutrient to use in fermentation media (Chapter 4). The marketable by-product helps to offset the cost of the treatment process. It is now recognized that water is no longer a cheap raw material (Chapters 4 and 12), hence there are considerable advantages in reducing the quantities

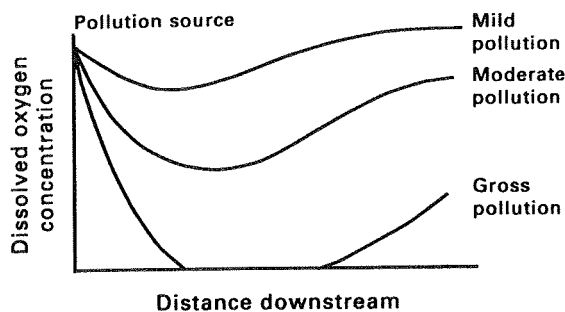


FIG. 11.1. The oxygen sag curve.

used and in recycling whenever it is feasible (Ashley, 1982). Obviously the introduction of good 'housekeeping' will lead to reductions in the volume of water used and the volume of effluent for treatment and final discharge. Recycling and reuse of materials, waste minimization, waste reduction at source and integrated pollution control are now very important factors to consider in the design and operation of any manufacturing facility, and may be the subject of new legislation in this field (Laing, 1992; Donaldson, 1993; McLeod and O'Hara, 1993).

DISSOLVED OXYGEN CONCENTRATION AS AN INDICATOR OF WATER QUALITY

Since oxygen is essential for the survival of most macro-organisms, it is important to ensure that there are adequate levels of dissolved oxygen in rivers, lakes, reservoirs, etc., if they are to be managed satisfactorily. Ideally, the oxygen concentration should be at least 90% of the saturation concentration at the ambient temperature and salinity of the water. It is therefore important to know how effluents containing soluble and particulate organic matter can influence the dissolved oxygen concentration. One widely used method of assessment is the 'biochemical oxygen demand' (BOD), which is a measure of the quantity of oxygen required for the oxidation of organic matter in water, by micro-organisms present, in a given time interval at a given temperature. The oxygen concentration of the effluent, or a dilution of it, is determined before and after incubation in the dark at 20° for 5 days. The oxygen decrease can then be determined titrimetrically and the results presented as mg of oxygen consumed per dm³ of sample. Mineral nutrients and a suitable bacterial inoculum are usually added to the initial sample to ensure optimal growth conditions. This test

is only an estimate of biodegradable material, hence recalcitrant or inhibitory compounds might be overlooked (SCA, 1989).

Because the BOD test takes 5 days it may be necessary to resort to the 'chemical oxygen demand' (COD), a chemical test which only takes a few hours to complete. The test is based on treating the sample with a known amount of boiling acidic potassium dichromate solution for 2.5 to 4 hours and then titrating the excess dichromate with ferrous sulphate or ferrous ammonium sulphate (HMSO, 1972). The oxidized organic matter is taken as being proportional to the potassium dichromate utilized. Most compounds are oxidized virtually to completion in this test, including those which are not biodegradable. In circumstances where substances are toxic to micro-organisms, the COD test may be the only suitable method available for assessing the degree of treatment required. The BOD:COD ratios for sewage are normally between 0.2:1 and 0.5:1. The ratio values for domestic sewage may be fairly steady. When industrial effluents of variable composition and loading are discharged, the ratio may fluctuate considerably. Very low BOD:COD ratios will indicate high concentrations of non biodegradable organic matter and consequently biological effluent treatment processes may be ineffective (Ballinger and Lishka, 1962; Davis, 1971). A number of alternative tests are available to indicate the 'oxygen demand' of a wastewater, including total organic carbon (TOC) and permanganate value (HMSO, 1972; American Public Health Association, 1992).

SITE SURVEYS

A complete survey of industrial operations is essential for any individual site before an economical waste-treatment programme can be planned. It is desirable to divide the facility into as many units as possible, as knowledge of the various material streams may show unexpected losses of finished product, solvent wastage, excessive use of water or unnecessary contamination of water which might be recycled, recovered or reused within the site. The factors, and concentrations where appropriate, listed in Table 11.1 ought to be known at all production rates under which an individual unit may operate in a representative time period.

The survey may indicate a need for better control of water usage and should identify sources of uncontaminated and contaminated water that might be reused in the factory. Concentrated waste streams should be kept separate if they contain materials that can be profitably

recovered. It is also often more economical to treat a concentrate rather than a large volume of a dilute effluent because of the saving on pumps and settling tank capacities, provided that concentrations do not reach toxic or inhibitory levels in biological treatment processes.

The various wastes may be tested in a laboratory and on a pilot scale to assess the best potential methods of chemical and biological treatment. Once the pHs of the effluents are known, samples may be mixed to see if a neutral pH is reached. A variety of tests may be used to establish methods for reducing salt concentrations, coagulating suspended particles and colloidal materials, and for breaking emulsions.

The commonly used biological tests include respirometry, aeration-flask tests (Otto *et al.*, 1962) and continuous-culture experiments. Small flask respirometers (Warburg or Gilson) and oxygen electrodes are used initially to establish the conditions to use in bio-oxidation of the effluent, and to test for the presence of toxic materials. Large respirometers (Simpson and Anderson, 1967) are useful for predicting effluent treatment rates and oxygen requirements. The residues in the flasks can be analysed to see if there are any recalcitrant materials. The use of laboratory continuous-culture vessels fitted with sludge-return pumps and settling tanks can provide detailed information (Ramathan and Gaudy, 1969). Proposed large-scale operating conditions for feed and aeration rates can be tested and their effectiveness assessed. The results from all these experiments may help in the design of a full-scale plant.

If the survey is comprehensive it should be possible to plan an overall treatment programme for a site and to establish:

1. Water sources which can be combined or reused.

2. Concentrated waste streams which contain valuable wastes to be recovered as food, animal feed, fertilizer or fuel.
3. Toxic effluents needing special treatment, or acids or alkalis needing neutralization.
4. The effluent loading expected under maximum production conditions.
5. The effluent(s) which might be discharged directly, without treatment, on to land or to a watercourse and not cause any pollution.
6. The effluent(s) which might be discharged into municipal sewers.

When all the relevant information has been obtained one can predict the size and type of effluent treatment plant required, and thus its capital and operational costs. This can then be compared with water company charges to treat the waste at an STW with and without on-site treatment. It should be remembered that the water company may insist on on-site treatment before a waste is discharged to the sewer, and will in most cases set consent limits for maximum flow rates and concentrations of specific analytes.

THE STRENGTHS OF FERMENTATION EFFLUENTS

It is already evident from earlier sections of this chapter that the presence of high levels of particulate or soluble organic matter in water will result in potential high BODs. This is precisely what is being achieved in all large-scale fermentation processes. An initial medium rich in organic matter is converted to biomass and primary and secondary metabolites. Unfortunately, the product often represents a small proportion of the initial raw material, even in an efficiently operated

TABLE 11.1. *Factors to investigate in a site survey*

Daily flow rate
Fluctuations in daily, weekly and seasonal flow
BOD/COD
Suspended solids
Turbidity
pH range
Temperature range
Odours and tastes
Colour
Hardness
Detergents
Radioactivity
Presence of specific toxins or inhibitors (e.g. heavy metals, phenolics etc.)

fermentation. The spent wastes remaining after the distillation of whisky may account for 90% of the initial raw organic materials, while in an antibiotic fermentation the effluent may represent in excess of 95%.

Data for a variety of fermentation effluents are summarized in Table 11.2. The BODs of many of these samples are much higher than that of domestic sewage and some may be comparable with strong effluents such as sulphite paper mill liquor. It is evident from these data that fermentation effluents may present serious potential pollution problems and may be expensive to dispose of unless well planned processes are used. A number of steps may be taken to reduce BODs in a process. Some of these will be discussed in this chapter. Careful selection of raw materials may have a significant effect on the type and quantities of effluent being produced. The cheapest raw material which meets the nutrient requirements of the micro-organisms may not be ideal if product yield, recovery cost, effluent disposal cost and possible by-product value are considered together. The high BOD value of fungal mycelium (40,000 to 70,000 mg dm⁻³) would indicate that any biomass should normally be kept separate from the remainder of an effluent and some of it may be sold as a by-product. It may also be worthwhile to

concentrate liquid fractions, for example, industrial alcohol and distillery stillages (10,000 to 25,000 mg dm⁻³) will both produce dried solubles fractions which can be sold.

Metabolites or components of some fermentation effluents may be extremely toxic and polluting and will require complete destruction, for example by chemical or thermal methods, before disposal. The need for such a treatment strategy will therefore make a significant contribution to the overall cost of the process. One such metabolite is avermectin produced by *Streptomyces avermitilis* fermentations. Here all effluent streams from the process are captured and any avermectin present chemically degraded (Omstead *et al.*, 1989).

TREATMENT AND DISPOSAL OF EFFLUENTS

The effluent disposal procedure which is finally adopted by a particular manufacturer is obviously determined by a number of factors, of which the most important is the control exercised by the relevant authorities in many countries on the quantity and quality of the waste discharge and the way in which it might be

TABLE 11.2. BOD strengths of effluents (mg dm⁻³)

Effluent	BOD	Reference
Domestic sewage	350	Boruff (1953)
Sulphite liquor from paper mill	20,000–40,000	
Beer:		
(a) spent grain press	15,000	Abson and Todhunter (1967)
(b) hop-press liquor	7430	
(c) yeast wash water	7400	
(d) spoil beer	up to 100,000	
(e) bottle washings	550	
Maltings:		
(a) suspended solids	1240	Koziorowski and Kucharski (1972)
(b) wastes	20–204	
(c) grain washings	1500	
Brewery effluent	1,400–1,800	Fang <i>et al.</i> (1990)
Industrial alcohol stillage	10,000–25,000	Blaine (1965)
Distillery stillage	10,000–25,000	Jackson (1960)
Yeast production	3,000–14,000	Boruff (1953)
Antibiotic waste	5,000–30,000	Jackson (1960)
Penicillin:		
(a) wet mycelium from filter	40,000–70,000	Boruff (1953)
(b) filtrate	2,150–10,000	
(c) wash water	210–13,800	
Streptomycin spent liquor	2,450–5,900	Koziorowski and Kucharski (1972)
Aureomycin spent liquor	4,000–7,000	
Solvents	up to 2,000,000	

done (Fisher, 1977). The range of effluent-disposal methods which can be considered is:

1. The effluent is discharged to land, river or sea in an untreated state.
2. The effluent is removed and disposed of in a landfill site or is incinerated.
3. The effluent is partially treated on site (e.g. by lagooning) prior to further treatment or disposal by one of the other routes indicated.
4. Part of the effluent is untreated and discharged as in 1 or 2, the remainder is treated at a sewage works or at the site before discharge.
5. All of the effluent is sent to the sewage works for treatment, although there might be reluctance by the sewage works to accept it, possibly resulting in some preliminary on-site treatment being required, and discharge rates and effluent composition defined.
6. All the effluent is treated at the factory before discharge.

DISPOSAL

Seas and rivers

The simplest way of disposal will be on a sea coast or in a large estuary where the effluent is discharged through a pipeline (installed by the factory or local authorities) extending below the low-water mark. In such a case there may be little preliminary treatment and one relies solely on the degree of dilution in the sea water.

If effluents are to be discharged into a river they must meet the requirements of the local river or drainage authorities. In Britain there is a Royal Commission standard requiring a maximum BOD (5 days) of 20 mg dm^{-3} and 30 mg dm^{-3} of suspended solid matter (the 20:30 standard). Stricter standards are often applied, depending on the use of the receiving water, such as a 10:10 standard; in addition, levels of ammoniacal nitrogen may be stipulated. There are, as well, often stringent upper limits for toxic metals and chemicals which might kill the fauna (particularly fish) and flora, e.g. sulphites, cyanides, phenols, copper, zinc, cadmium, arsenic, etc. It is highly unlikely that one would be able to discharge an industrial waste today without some form of pretreatment.

Lagoons (oxidation ponds)

Lagoons, holding ponds, oxidation ponds, etc., may be used by a number of industries if land is available at a reasonable cost. It is a method often used in seasonal industries where capital investment in effluent plant is difficult to justify. The lagoon normally consists of a volume of shallow water enclosed by watertight embankments. Oxidation ponds are typically 1–2 m deep. They can be designed to maintain aerobic conditions throughout, but more commonly decomposition at the surface is aerobic and that nearer the bottom is anaerobic and they are then known as facultative ponds. Oxygen for aerobic degradation is provided both from the surface of the pond, and from algal photosynthesis. Deeper ponds (known as lagoons) are mechanically agitated to provide aeration. Lagoons are simple to build and operate, but are expensive in terms of land requirements. They may be used as the sole method of treatment, incorporating both physical (sedimentation) and biological processes, but the effluent produced may not reach locally acceptable standards. Alternatively they can provide an initial pretreatment or can be used to 'polish' effluent from secondary treatment processes.

Spray irrigation

Liquid wastes can be applied directly to land as irrigation water and fertilizer when they are claimed to have a number of beneficial effects on the soil and plants. If this method of disposal is to be used, then it is necessary to have a large area of land near to a manufacturing plant in an area of low to medium rainfall. Pipeline costs will often restrict use of this technique. Colovos and Tinklenberg (1962) described the disposal of antibiotic and steroid wastes with BODs of 5000 to $20,000 \text{ mg dm}^{-3}$. These wastes were initially chlorinated to lower the BOD and reduce unpleasant odours and then sprayed on to land until the equivalent of 38 mm of rainfall was reached. This process was repeated at monthly intervals and improved plant growth.

When appropriate, solid wastes may be spread onto land as a fertilizer and soil conditioner. This practice is common with sewage sludges, and Mbagwu and Ekwelor (1990) report the use of spent brewers grains to improve the productivity of fragile soils. Irrespective of whether the waste is liquid or solid, the concentration of heavy metals and certain organic components will require careful monitoring and control to safeguard the environment and public health.

Well disposal

Disused wells, boreholes or mine shafts may provide an ideal, cheap method for disposal when the volume of waste is limited, the underground strata are suitable and the chances of contamination of water supplies utilized by water authorities are negligible (Zajic, 1971). Melcher (1962) has described the use of wells 500-m deep for the daily disposal of:

Acetic acid	900 kg
Ammonium acetate	900 kg
Sodium acetate	760 kg
Sodium chloride	450 kg
Sodium and ammonium bromide	225 kg
Methanol, xylene, tars and organic compounds	

This mixture had a pH of 4 to 5 and a COD of 40,000 to 60,000 mg dm⁻³ rising to 100,000 mg dm⁻³ and was pumped into the wells at 50 to 100 dm³ min⁻¹.

Careful hydrogeological surveys will be needed to prove that waste disposal in wells will not cause pollution of aquifers and threaten groundwater supplies.

Landfilling

Landfilling is a disposal method for municipal solid waste (MSW) and industrial waste. It utilizes natural or man made voids (e.g. disused clay pits) into which the waste is deposited. Both solid and liquid wastes can be deposited depending on restrictions imposed by the site licence. Strict controls exist on the amount of liquid and toxic materials which can be accepted because of the threat of groundwater pollution if leachate (a liquid having BOD levels up to 30,000 mg dm⁻³) escapes from the site. Leachate is generated from liquid deposited in the site, water entering the site naturally via precipitation or surface run-off and by anaerobic microbial action as organic matter in the landfill is degraded. Microbial action similar to that in anaerobic digesters leads to the production of landfill gas (LFG) which, being 50–60% methane can, if collected efficiently, provide a useful source of energy (Freestone *et al.*, 1994).

Incineration

A number of designs exist for the incineration of solid and/or liquid wastes either on site or at a

commercial incinerator, including rotary kilns, fluidized beds and multiple hearth furnaces. Combustion temperatures need to be carefully controlled to destroy and prevent the formation of dioxins and furans, formation of which occurs at between 300° and 800°, and total destruction is effected at temperatures above 1000° with a retention time of 1 second. Flue gases from the incinerator require cleaning to remove particulates, acid vapours, etc. using electrostatic precipitators, cyclones and wet scrubbers to comply with local environmental protection standards. Waste disposal by incineration is currently significantly more expensive than landfilling, with costs for the disposal of MSW being \$37 tonne⁻¹ for incineration compared with \$10 tonne⁻¹ for landfilling (Smith, 1993).

Disposal of effluents to sewers

Municipal authorities and water treatment companies which accept trade effluents into their sewage systems will want to be sure that:

1. The sewage works has the capacity to cope with the estimated volume of effluent.
2. The effluent will not interfere with the treatment processes used at the sewage works.
3. There are no compounds present in the effluent which will pass through the sewage works unchanged and then cause problems when discharged into a watercourse.

It is common practice for local authorities to demand preliminary on-site pretreatment before discharge into sewers to minimize the effects of industrial wastes. The actual pretreatment required will depend on the precise nature of the waste and may range from simple sedimentation to complex chemical and biological processes.

TREATMENT PROCESSES

Fermentation wastes may be treated on-site or at an STW by any or all of the three following methods:

1. Physical treatment.
2. Chemical treatment.
3. Biological treatment.

The final choice of treatment and disposal processes